

# Release 1 Document AI Pipeline and Evaluation Specification

**Provider research snapshot:** August 1, 2026

## 1. Executive pipeline summary

Release 1 should use a **text-first hybrid Document AI pipeline**. The worker first inspects the PDF and extracts embedded text while preserving page boundaries and positional information. Pages with inadequate text are selectively converted and processed through OCR. The resulting page-aware text becomes the main extraction context. Selected page images may be added only when tables, layout, scan quality, or ambiguous associations cannot be understood reliably from text alone. One structured AI extraction call produces supported candidate values and proposed evidence. Deterministic processing then validates structure, normalizes values, verifies evidence against source artifacts, detects conflicts, and creates explainable warnings. A candidate enters human review only after passing a publication gate. It never updates approved policy information directly.

**One-sentence pipeline definition:** A page-aware, text-first pipeline that uses selective OCR and limited visual context to produce validated, evidence-backed candidate policy data for human review.

### Recommended high-level strategy

1. Inspect and validate the source PDF.
2. Extract direct text and page positions.
3. Evaluate text quality per page.
4. OCR only pages that need it.
5. classify the document using deterministic signals first.
6. Detect several-policy risk.
7. Prepare page-aware text and selected images.
8. Run one structured extraction call.
9. Allow one controlled structural-repair attempt only when necessary.
10. Normalize and deterministically validate values.
11. Verify evidence against source content.
12. Generate explainable warnings.
13. Publish an immutable candidate for review.
14. Preserve all attempt versions and usage details.

### Provisional provider direction

The provisional configuration to benchmark first is:

- **Direct parser:** PyMuPDF
- **OCR baseline:** OCRmyPDF with Tesseract
- **Managed OCR challenger:** Google Enterprise Document OCR
- **Primary AI benchmark candidate:** Gemini 3.6 Flash
- **Cost-focused AI challenger:** OpenAI GPT-5.6 Luna
- **Reserve difficult-document candidate:** Claude Sonnet 5
- **Input representation:** page-aware text plus selected page images

- **Extraction structure:** one structured extraction call followed by deterministic validation
- **Evidence:** model-proposed page evidence verified against parser or OCR output

These are evaluation candidates, not approved production winners.

## Required benchmark gate

No parser, OCR approach, model, or input strategy becomes final until it passes:

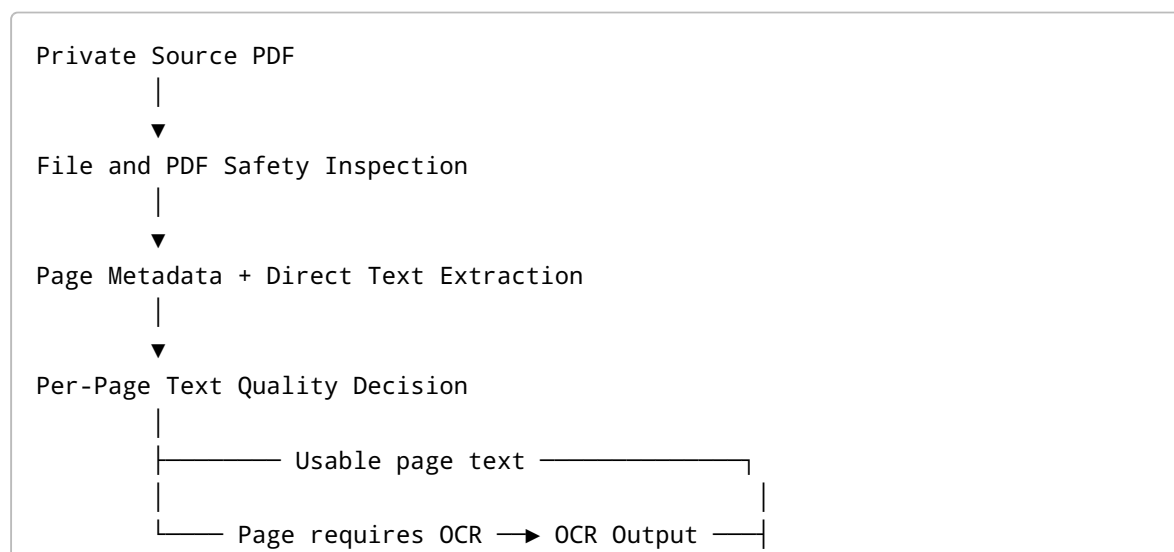
- supported and unsupported document tests
- clean and scanned field-accuracy targets
- critical-field silent-error rules
- vehicle and coverage association tests
- evidence-support tests
- repeatability tests
- latency limits
- cost limits approved by the product owner
- provider-failure and malformed-output tests

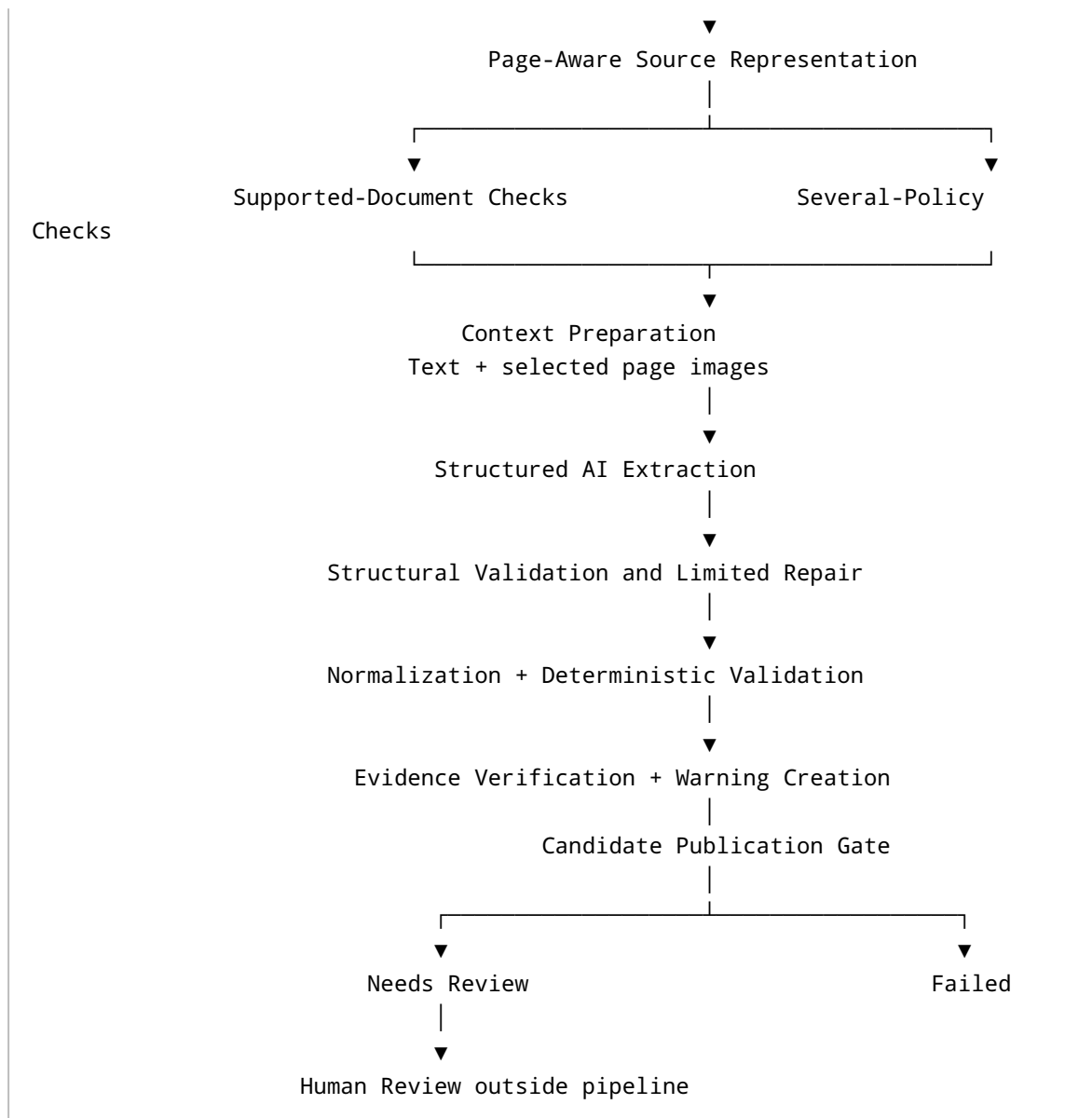
## Complexity intentionally avoided

Release 1 does not use:

- several production models
- automatic provider routing
- a verification model on every document
- model fine-tuning
- automatic learning from reviewer corrections
- OCR on every page
- per-page AI calls by default
- autonomous approval
- general document schemas
- carrier-specific extraction systems
- a generalized workflow engine

## Pipeline diagram





## 2. Pipeline goals and non-goals

### 2.1 Goals

#### Accurate supported-field extraction

Extract only the approved policy, named-insured, vehicle, driver, and coverage fields.

#### Explainable evidence

Every important candidate value should link to:

- a source page
- a supporting excerpt
- source type

- exact location when reliable

### **No invented values**

Information that is not supported remains missing, unclear, or conflicting.

### **Relationship preservation**

The pipeline must preserve:

- individual vehicles
- individual drivers
- vehicle-specific deductibles
- policy-level liability coverages
- primary and additional named insureds

### **Reliable warnings**

Warnings should arise from observable evidence, validation, conflicts, OCR conditions, and associations.

### **Safe scan handling**

Readable scanned PDFs should produce reviewable results without pretending OCR is exact.

### **Repeatable benchmark behavior**

A fixed pipeline version should produce acceptably stable values, evidence, lists, and warning states.

### **Controlled latency and cost**

Direct text and cached artifacts should be reused. OCR and AI calls occur only when justified.

### **Provider replaceability**

Provider-specific behavior remains behind narrow boundaries.

### **Stored Result compatibility**

Stored and live candidates must use the same canonical contract and review workflow.

### **Human-review readiness**

The pipeline output must provide enough evidence and warnings for a reviewer to decide safely.

### **Regression detection**

Every material parser, OCR, model, instruction, normalization, validation, or evidence change must be measurable against a fixed corpus.

## 2.2 Non-goals

Release 1 does not provide:

- autonomous approval
  - quoting
  - policy verification
  - carrier verification
  - coverage recommendations
  - general document extraction
  - arbitrary extraction fields
  - handwriting recognition
  - multilingual extraction
  - automatic retraining
  - automatic prompt changes
  - multiple production AI providers
  - automatic provider fallback
  - perfect support for every carrier or state layout
  - public arbitrary document uploads
  - model confidence percentages as correctness proof
  - coverage adequacy analysis
  - fraud detection
- 

## 3. Pipeline constraints inherited from approved documents

### 3.1 Supported input

- English-language U.S. personal auto declaration pages
- PDF only
- one PDF at a time
- one policy per file
- maximum 15 MB
- maximum 10 pages
- selectable-text PDFs
- readable scanned PDFs

### 3.2 Supported fields

Only the approved fields may be extracted. Unexpected fields must be ignored or reported diagnostically, not added to the product contract.

### 3.3 Direct text first

Embedded text is attempted before OCR.

### 3.4 Selective OCR

OCR is used only when direct text is missing, corrupt, incomplete, or unsuitable.

### **3.5 One production provider per capability**

One parser strategy, one production OCR approach, and one production AI model are selected after evaluation.

### **3.6 Background execution**

All slow processing runs in a worker and survives browser closure.

### **3.7 Preserved attempts**

Each attempt retains its own:

- parser result
- OCR result
- model result
- candidate
- validation
- evidence
- warnings
- usage
- pipeline version

### **3.8 Source evidence**

Critical and important values require meaningful source support.

### **3.9 Warning severity**

The pipeline must produce Information, Important, and Critical warning candidates that follow the approved review behavior.

### **3.10 Human approval boundary**

Pipeline output stops at a candidate result. It cannot create approved policy information.

### **3.11 Candidate immutability**

The normalized candidate produced by one attempt is immutable. Reviewer edits live in Review, not Extraction Results.

### **3.12 Retry limits**

No more than two user-triggered Technical Retries follow the initial attempt.

### **3.13 Reprocessing**

A new pipeline configuration or intentional rerun creates another preserved attempt.

### 3.14 Stored Result replay

Stored artifacts enter the same candidate-publication and review contract without pretending to be live.

### 3.15 Raw-response retention

Raw provider responses use limited configurable retention, except approved evaluation samples.

### 3.16 Privacy and logging

Normal logs must not contain:

- document text
  - names
  - addresses
  - full policy numbers
  - full VINs
  - raw prompts
  - raw provider responses
  - signed document URLs
- 

## 4. Recommended pipeline shape

### 4.1 Strategy A: Direct text, OCR fallback, text-based AI extraction

#### Description

Extract page-aware text. OCR only deficient pages. Send text to the AI model.

#### Strengths

- lowest typical cost
- good reproducibility
- easy page evidence
- strong privacy control over selected content
- provider-neutral
- easy stage caching
- predictable prompt size

#### Weaknesses

- tables and multi-column layouts may lose relationships
- visual grouping may be lost
- OCR ordering can be imperfect

### 4.2 Strategy B: Direct multimodal processing of full PDF pages

#### Description

Send the PDF or all page images directly to a multimodal model.

## **Strengths**

- preserves layout
- handles visual tables
- simpler first prototype
- may reduce custom OCR work

## **Weaknesses**

- higher and less predictable token use
- stronger provider dependency
- weaker artifact-level reproducibility
- harder evidence verification
- full document is transmitted
- can hide whether errors came from reading or reasoning

OpenAI's current PDF-input behavior supplies both extracted text and page images to vision-capable models, which improves visual understanding but increases token use. Anthropic and Google also support native PDF or document understanding. These capabilities are useful benchmark alternatives, but native support alone does not prove better extraction quality for this workflow.

## **4.3 Strategy C: Hybrid text plus selected page images**

### **Description**

Use direct or OCR text as the primary context. Add images only for selected pages with tables, poor text, or ambiguous associations.

### **Strengths**

- retains most cost and reproducibility benefits of text
- provides visual help where needed
- limits document transmission
- supports difficult tables
- preserves stage-level diagnosis

### **Weaknesses**

- requires page-selection logic
- multimodal input increases model complexity
- benchmark must prove page selection does not omit needed visual context

## **4.4 Strategy D: Per-page extraction plus aggregation**

### **Description**

Extract each page independently and combine results afterward.

### **Strengths**

- handles long documents



- isolates page failures
- page evidence is natural

### **Weaknesses**

- several AI calls
- repeated policy context
- duplicate candidates
- difficult cross-page associations
- higher latency and cost
- aggregation may create new errors

## **4.5 Recommendation**

Use **Strategy C** as a controlled extension of **Strategy A**:

Page-aware direct text, selective OCR, one document-level structured extraction call, and selected page images only where benchmark evidence shows visual context is needed.

## **4.6 Simpler alternative**

Pure text-only Strategy A is simpler and should be the first benchmark baseline.

## **4.7 Why additional complexity may be justified**

Declaration pages often contain:

- coverage tables
- vehicles spread across columns
- values associated by visual row
- scanned tables with weak reading order

Selected page images are justified only if text-only extraction fails association or evidence targets.

## **4.8 Accepted trade-off**

The pipeline needs page-image preparation and context-selection logic, but avoids sending every page image by default.

## **4.9 Reconsideration trigger**

Use text-only production extraction if it meets all release gates. Use full native-PDF multimodal processing only if it materially improves critical accuracy and association quality without violating cost and repeatability limits.

## 5. End-to-end processing stages

| Stage                       | Purpose and input                                            | Conceptual output                     | Main failures                               | Paid?   | Preserve? | Replayable? | Storage   |
|-----------------------------|--------------------------------------------------------------|---------------------------------------|---------------------------------------------|---------|-----------|-------------|-----------|
| 1. Acceptance confirmation  | Confirm authorized stored file, customer, limits and attempt | Accepted source reference             | Missing upload, authorization mismatch      | No      | Yes       | Yes         | Yes       |
| 2. Secure file inspection   | Verify real PDF content and safe basic structure             | Safe-to-open decision                 | Wrong type, malformed, encrypted            | No      | Summary   | Yes         | Usually   |
| 3. PDF metadata inspection  | Read pages, dimensions and document metadata                 | Page inventory                        | Zero pages, unreadable tree, parser timeout | No      | Yes       | Yes         | Yes       |
| 4. Direct text extraction   | Read embedded text page by page                              | Page text, blocks, positions          | Encoding failure, parser failure            | No      | Yes       | Yes         | Sometimes |
| 5. Text-quality decision    | Decide whether each page's text is usable                    | Per-page source decision              | Unexplainable or inconsistent quality       | No      | Yes       | Yes         | No        |
| 6. Orientation handling     | Detect and normalize page orientation for OCR                | Rotation metadata and prepared page   | Orientation cannot be resolved              | No      | Yes       | Yes         | Rarely    |
| 7. OCR fallback             | Read selected scanned or deficient pages                     | OCR text, positions, quality metadata | OCR timeout, unusable text                  | Usually | Yes       | Yes         | Sometimes |
| 8. Document-type evaluation | Determine supported, unsupported or ambiguous                | Classification result and evidence    | Insufficient readable content               | Maybe   | Yes       | Yes         | Sometimes |

| Stage                     | Purpose and input                                 | Conceptual output                 | Main failures                              | Paid?          | Preserve?         | Replayable? | Storage        |
|---------------------------|---------------------------------------------------|-----------------------------------|--------------------------------------------|----------------|-------------------|-------------|----------------|
| 9. One-policy evaluation  | Detect multiple policy identities                 | One-policy, ambiguous or multiple | Several inseparable policies               | No or included | Yes               | Yes         | Sometimes      |
| 10. Context preparation   | Build page-aware model input                      | Reproducible extraction context   | Context too large, missing page references | No             | Version reference | Yes         | Yes            |
| 11. Structured extraction | Extract approved fields and proposed evidence     | Raw structured result             | Timeout, refusal, malformed output         | Yes            | Limited raw       | Yes         | No immediate   |
| 12. Structural validation | Check contract shape and relationships            | Valid or repairable structure     | Missing sections, invalid lists            | No             | Yes               | Yes         | Sometimes      |
| 13. Normalization         | Produce canonical candidate forms                 | Normalized candidate              | Ambiguous transformation                   | No             | Yes               | Yes         | No             |
| 14. Business validation   | Detect invalid dates, identities and associations | Validation signals                | Internal rule/config mismatch              | No             | Yes               | Yes         | Sometimes      |
| 15. Evidence preparation  | Verify pages and supporting excerpts              | Evidence records                  | Unsupported excerpt, missing page          | No             | Yes               | Yes         | Sometimes      |
| 16. Warning generation    | Convert signals into explainable review warnings  | Warning set                       | Configuration mismatch                     | No             | Yes               | Yes         | Sometimes      |
| 17. Candidate publication | Apply publication gate                            | Immutable candidate or failure    | Fatal blockers                             | No             | Yes               | Yes         | Final decision |

| Stage                  | Purpose and input                 | Conceptual output      | Main failures                   | Paid? | Preserve? | Replayable? | Storage  |
|------------------------|-----------------------------------|------------------------|---------------------------------|-------|-----------|-------------|----------|
| 18. Usage finalization | Record actual usage and estimate  | Final usage record     | Missing provider usage metadata | No    | Yes       | Yes         | Metadata |
| 19. Attempt completion | Mark attempt reviewable or failed | Needs Review or Failed | Transaction failure             | No    | Yes       | Yes         | Final    |

## Stage preservation principle

Preserve the smallest artifact needed to:

- reproduce the candidate
- explain the result
- avoid repeated paid calls
- compare pipeline versions
- diagnose failure

Do not preserve every temporary rendering product indefinitely.

## 6. Input preflight and PDF safety

### 6.1 Rejection before a lasting processing attempt

Reject before paid processing when the product can safely determine:

- file content is not PDF
- file exceeds 15 MB
- page count exceeds 10
- password or unsupported encryption is present
- PDF contains zero pages
- basic structure cannot be opened
- upload authorization does not match the stored object

The product may still preserve a minimal upload-failure audit event.

### 6.2 Failed processing attempt

Create a Failed attempt when the file passed initial acceptance but deeper processing discovers:

- unreadable page tree
- corrupt internal objects
- parser timeout
- resource exhaustion
- no usable visible pages

- unsupported document content
- OCR failure
- processing configuration mismatch

### **6.3 Replacement-required failures**

Replacement is required for:

- corrupted PDF
- password-protected PDF
- severely unreadable scan
- zero readable pages
- several inseparable policies
- unsupported document
- excessive page count
- wrong source file

### **6.4 Retryable technical failures**

May include:

- temporary storage access failure
- parser process interruption
- OCR service timeout
- AI provider timeout
- temporary provider unavailability
- worker interruption
- transient rendering failure

### **6.5 Embedded attachments**

Ignore embedded attachments unless they are necessary to render the main document. Do not extract or execute them.

### **6.6 Embedded scripts**

Never execute PDF JavaScript or active content.

### **6.7 Unsafe filenames**

Use the filename only as a user-visible label. Internal storage identity must not depend on it.

### **6.8 Large page dimensions**

Apply rendering limits. A PDF under 15 MB may still contain pages that require unreasonable memory when rasterized.

## 6.9 Resource protection

Each parser and renderer stage requires:

- maximum execution time
- memory budget
- controlled rendering resolution
- maximum image dimensions
- isolated worker execution
- cleanup after failure

## 6.10 Duplicate content

Calculate content identity from file bytes. File name, upload date, or customer name alone is insufficient.

## 6.11 Malware scanning

### Public portfolio demo

No general malware-scanning platform is required because users cannot upload arbitrary files.

### Owner-controlled development

Strict parser isolation and curated documents are sufficient initially.

### Before real customer uploads

Add:

- malware scanning or managed file scanning
- quarantine before deeper processing
- PDF dependency security review
- file-retention and deletion procedures
- provider data-handling review

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# 7. Direct PDF text extraction specification

## 7.1 Required output

Direct extraction should produce for every page:

- page number
- plain text
- text blocks
- word-level text where practical
- positional information
- page dimensions
- detected rotation
- extraction warnings

- source-order metadata

## 7.2 Page boundaries

Page boundaries must never be removed. Every text block and later evidence excerpt must retain its source page.

## 7.3 Reading order

The parser may return text in drawing order rather than human reading order.

The pipeline should retain:

- parser-native block order
- position information
- a normalized reading-order representation

It must not destroy the original block sequence.

## 7.4 Repeated headers and footers

Repeated carrier names, page labels, confidentiality messages, and page numbers may be marked as repeated context. They should not automatically be removed before evidence analysis.

## 7.5 Tables

Text extraction may flatten rows and columns. Positional blocks and words must be retained so table relationships can be reconstructed or reviewed.

## 7.6 Font encoding

Some PDFs visually display correct characters but expose unusable embedded text. Character quality checks must catch this before trusting direct text.

## 7.7 Hidden or duplicate text

Scanned PDFs may include an invisible OCR layer and visible page image. The pipeline should detect:

- repeated text
- text far outside visible page bounds
- duplicated OCR layers
- conflicting visible and embedded content

## 7.8 Parser shortlist

### PyMuPDF

PyMuPDF supports page text extraction as plain text, blocks, words, and positional information. It also supports page rendering, making it useful for evidence, page images, and selective OCR preparation.

### **Strengths**

- fast
- word and block positions
- page rendering
- one library for extraction and image preparation
- useful evidence support

### **Limitations**

- reading order still needs validation
- licensing and packaging must be reviewed before commercial deployment
- parser behavior may differ across complex PDFs

### **pypdf**

pypdf is a pure-Python PDF library with text and metadata extraction, including a layout-oriented extraction mode. Its documentation also emphasizes that digitally generated PDFs contain information OCR may lose.

### **Strengths**

- pure Python
- mature PDF manipulation
- simple metadata access
- useful independent fallback during evaluation

### **Limitations**

- less suited to page rendering
- weaker fit for positional evidence and OCR preparation
- not designed as an OCR engine

## **7.9 Provisional recommendation**

Use **PyMuPDF** as the **primary parser**.

Use pypdf only as:

- an evaluation comparison
- a fallback metadata inspection option if PyMuPDF encounters a specific unsupported case

Do not run two parsers on every production document.

## **7.10 Accepted trade-off**

PyMuPDF creates a stronger dependency, but it reduces the need for a separate page-rendering library.



## 7.11 Benchmark reversal condition

Reconsider if PyMuPDF:

- fails materially more benchmark PDFs
  - produces unreliable positional evidence
  - creates licensing constraints
  - causes unacceptable worker memory or deployment issues
- 

## 8. Text-quality decision

### 8.1 Explainable signals

Evaluate each page using observable signals.

#### Text quantity

- characters
- words
- text blocks
- proportion of nearly empty pages

#### Character quality

- replacement characters
- control characters
- repeated nonsensical symbols
- unusually high punctuation ratio
- broken word fragments

#### Layout quality

- words in implausible order
- overlapping duplicate text
- severe column mixing
- very large gaps between visual content and extracted text

#### Insurance-context signals

Presence or absence of likely labels such as:

- policy
- effective
- expiration
- insured
- vehicle
- coverage
- deductible

These terms are supporting signals, not proof of document type.

### **Visual-text mismatch**

- page appears image-heavy but has little text
- extracted text covers only a small visible region
- invisible text does not align with page positions

## **8.2 Outcomes**

### **Direct text sufficient**

Use direct text as the primary page source.

### **Selected pages require OCR**

Use OCR only for deficient pages.

### **Full document requires OCR**

Use when almost all pages are image-only or embedded text is broadly corrupted.

### **Too poor to continue**

Fail when neither direct text nor OCR can provide reviewable source content.

### **Continue with quality warning**

Allowed when enough information exists for human review but evidence quality is limited.

## **8.3 Threshold policy**

Initial thresholds must be provisional and tuned using the benchmark.

Examples of threshold categories:

- minimum meaningful characters per visually non-empty page
- maximum proportion of replacement characters
- maximum proportion of empty pages
- minimum alignment between words and page bounds
- minimum expected-term coverage for classification

## **8.4 Recommendation**

Use a **rule set with named reasons**, not one unexplained quality percentage.

## **8.5 Accepted trade-off**

Rules require benchmark tuning but produce understandable OCR decisions and diagnostics.

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## 9. OCR page-selection strategy

### 9.1 OCR every page

#### Advantage

Consistent source representation.

#### Problems

- unnecessary cost
- slower
- can degrade high-quality digital text
- may create OCR mistakes where exact text already exists

Rejected.

### 9.2 OCR only pages with no text

#### Advantage

Very simple.

#### Problem

Pages may contain broken or incomplete text that is not literally empty.

Insufficient.

### 9.3 OCR full document when any page fails

#### Advantage

One consistent OCR output.

#### Problems

- excessive cost
- discards better digital text
- duplicates content
- weak mixed-document handling

Rejected.

### 9.4 Hybrid page-by-page selection

#### Recommendation

Evaluate each page independently and OCR only pages whose direct text is inadequate.

## 9.5 Coexistence rules

Each page has one selected primary text representation:

- Direct Text
- OCR Text
- Direct Text with OCR Supplement

A supplement is permitted only when the two sources cover different visible regions and duplication can be controlled.

## 9.6 Ordering

Final context preserves original PDF page order regardless of text source.

## 9.7 Duplicate control

When OCR is run on a page with partial text:

- compare overlapping text regions
- choose one primary representation per block
- do not concatenate both entire page outputs blindly

## 9.8 Mixed PDFs

A six-page document may contain:

- pages 1–3 direct text
- page 4 OCR
- page 5 direct text
- page 6 OCR

The extraction context records the source type for every block.

## 9.9 Cost benefit

Selective OCR minimizes:

- OCR-page charges
  - worker CPU
  - image rendering
  - processing time
  - external document transmission
-

## 10. OCR approach and provider evaluation

### 10.1 Local baseline: OCRmyPDF and Tesseract

OCRmyPDF adds searchable OCR text to PDFs using Tesseract and supports orientation correction, deskewing, and selected image cleanup. Tesseract can emit hOCR data containing recognized text, bounding boxes, and confidence metadata.

#### Strengths

- no per-page provider charge
- local processing
- good repeatability
- bounding-box evidence
- straightforward development fixtures
- no external OCR transmission

#### Limitations

- worker CPU and memory
- weaker difficult-layout performance
- may require image preprocessing
- confidence is not correctness proof
- language and font variability

### 10.2 Managed challenger: Google Enterprise Document OCR

Google's current Enterprise Document OCR documentation describes text and layout extraction, image-quality analysis, page orientation handling, and structured document output. Current listed pricing begins at \$1.50 per 1,000 pages for the first five million pages, subject to Google's current pricing terms.

#### Strengths

- low listed page cost
- managed scaling
- layout and bounding-box information
- image-quality signals
- reduced local worker load
- credible scan-quality challenger

#### Limitations

- another external service
- credential and project configuration
- document transmission
- provider-specific response
- network latency and rate limits

### 10.3 AWS Textract

Credible for forms and tables, but adding a separate AWS dependency solely for OCR conflicts with the Release 1 platform-minimization goal unless it clearly wins the benchmark.

It should not be in the first full comparison unless local and Google OCR both fail.

### 10.4 Azure AI Document Intelligence

Also credible for OCR and layout, but creates another provider relationship. It should remain a reserve option rather than expanding the initial benchmark.

### 10.5 Direct multimodal AI reading

A multimodal model may read scanned pages directly.

This should be benchmarked as an **input-strategy alternative**, not treated as OCR infrastructure, because:

- evidence coordinates may be weaker
- text artifacts are harder to reuse
- cost depends on image processing
- reading and extraction errors become harder to separate

### 10.6 Provisional recommendation

Use:

- **OCRmyPDF/Tesseract as the provisional local OCR approach**
- **Google Enterprise Document OCR as the managed benchmark challenger**

### 10.7 Lower-cost fallback

The local approach is the fallback when:

- scan quality targets are met
- worker resource use is acceptable
- evidence boxes are sufficient

### 10.8 Benchmark promotion rule

Promote managed OCR when it improves one or more of the following materially:

- policy number accuracy
- VIN accuracy
- table row preservation
- evidence-page and location quality
- rotated scan handling
- OCR technical failure rate

without violating the approved cost ceiling.

## 10.9 Why one production OCR approach is enough

Maintaining two production OCR paths would multiply:

- warning behavior
- evidence formats
- failure handling
- regression testing
- provider configuration

The losing approach may remain a development benchmark, not a runtime fallback.

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## 11. Rotation and page-image handling

### 11.1 Rotation

Detect and record page rotation separately for each page.

Supported:

- 90 degrees
- 180 degrees
- 270 degrees
- mixed orientation

### 11.2 Automatic correction

Orientation should be corrected for OCR input when detection is reliable.

The original source PDF remains unchanged.

### 11.3 Viewer rotation

The reviewer can rotate the visual page independently. Viewer rotation does not alter processing artifacts.

### 11.4 Skew

Basic deskewing is permitted before OCR.

### 11.5 Low-resolution scans

Continue when OCR remains usable, but create OCR Concern warnings for affected fields.

### 11.6 Dark or washed-out scans

Basic contrast and cleanup may be benchmarked.

Advanced restoration is excluded.

## 11.7 Borders and shadows

Simple cropping or cleanup may occur when it improves OCR without removing content.

## 11.8 Duplicate pages

Detect strongly matching page images or text.

A duplicate-page warning may be informational unless it creates conflicting policy interpretation.

## 11.9 Required Release 1 behavior

- orientation detection
- OCR-oriented rotation
- basic deskew
- controlled rendering
- quality warning
- failure for unreadable pages

## 11.10 Deferred behavior

- advanced deblurring
  - handwriting enhancement
  - damaged-paper restoration
  - photo-perspective correction
  - manual image-editing tools
- 

# 12. Supported-document classification

## 12.1 Deterministic heuristics

Signals may include combinations of:

- personal auto terminology
- declaration terminology
- policy identity
- named insured
- vehicles
- liability coverages
- deductibles
- carrier identity
- effective period

### Benefit

Cheap and explainable.

### Limitation

Carrier layouts vary.



## **12.2 AI classification**

### **Benefit**

Handles semantic variation.

### **Limitation**

Adds cost and may confidently misclassify.

## **12.3 Extraction-first classification**

Attempt extraction and judge from output.

### **Benefit**

No separate classification call.

### **Limitation**

May spend AI cost on clearly unsupported documents.

## **12.4 Combined approach**

### **Recommendation**

Use deterministic checks first, then include document-type evaluation inside the single structured extraction call when deterministic evidence is ambiguous.

## **12.5 Outcomes**

### **Supported**

Enough evidence indicates a personal auto declaration page.

Proceed.

### **Unsupported**

Clear evidence indicates:

- insurance card
- home declaration
- commercial policy
- claim form
- invoice
- billing notice
- quote without declaration context
- unrelated document

Fail as Unsupported Document.

### **Ambiguous**

Continue only when:

- enough source text exists
- the AI extraction can explicitly report document-type ambiguity
- candidate publication can produce a Critical warning rather than fabricate data

### **12.6 Early stop**

Stop before AI extraction for obvious:

- non-insurance PDF
- insurance ID card
- invoice
- blank or unreadable document

### **12.7 Proceed to review**

An ambiguous insurance document may reach Needs Review with Critical warning when:

- it resembles a declaration page
- supported policy fields exist
- one human decision could resolve the ambiguity

### **12.8 No perfect-classification claim**

Classification performance must be reported separately, including false acceptance of unsupported documents.

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## **13. One-policy-per-file detection**

### **13.1 Signals**

Detect:

- several distinct policy numbers
- several non-overlapping effective periods
- several carriers
- repeated declaration title pages
- several primary named insured identities
- repeated page sets
- appended unrelated policy pages
- renewal and current declarations combined

## 13.2 Automatic failure

Fail as several inseparable policies when:

- two complete policy identities exist
- pages can clearly be divided into distinct policies
- merging would be unsafe
- no single policy can be selected without discarding major content

## 13.3 Critical review warning

Publish with Critical warning when:

- one likely primary policy exists
- a second policy number may be an old reference, renewal reference, or related identifier
- a reviewer can resolve the ambiguity from source evidence

## 13.4 Recommended reviewer outcome

For confirmed multiple policies:

- reject
- separate files
- re-upload one policy per PDF

## 13.5 Prohibited behavior

The pipeline must never:

- merge vehicles across policies
- choose the latest period silently
- combine coverage values
- treat a renewal and prior declaration as one policy
- discard extra policy pages without warning

---

# 14. Document context preparation

## 14.1 Full direct text

Useful when:

- page order is reliable
- tables remain understandable
- context fits comfortably

## 14.2 Cleaned page text

Recommended primary representation.

Cleaning may:

- normalize whitespace
- mark page boundaries
- identify repeated headers
- retain block positions
- retain source representation

Cleaning must not:

- remove repeated policy numbers automatically
- rewrite OCR characters
- infer table associations
- combine rows destructively

### **14.3 OCR text**

Used per selected page and clearly labelled as OCR-derived.

### **14.4 Selected text blocks**

Useful when the full document contains irrelevant boilerplate, but selection must not omit important coverage context.

### **14.5 Page images**

Use for:

- visually important tables
- pages with weak reading order
- ambiguous row associations
- selected scan-quality problems

### **14.6 Full PDF attachment**

Benchmark alternative only.

Not the default because it:

- transmits all content
- reduces stage control
- may increase token use
- makes provider behavior less transparent

### **14.7 Recommended context**

For every page:

- page marker
- selected primary text
- block or row markers when useful
- source type

- repeated-header annotation
- page image only when selected

## 14.8 Context-window handling

The 10-page limit should normally fit modern model contexts, but context still needs control because OCR and page images can be verbose.

If context is unexpectedly large:

1. remove only verified duplicate boilerplate
2. preserve policy and table regions
3. use selected page images
4. fail rather than silently truncate critical pages

## 14.9 Reproducibility

The exact prepared-context version or deterministic source artifacts must be identifiable for every attempt.

---

# 15. AI-call decomposition

## 15.1 One call for classification, extraction, and evidence

### Benefit

Lowest latency and cost.

### Risk

One malformed result affects all outputs.

## 15.2 Separate classification and extraction calls

### Benefit

Clear separation.

### Risk

Extra cost and conflicting AI judgments.

## 15.3 One extraction call plus deterministic validation

### Recommendation

Use one structured extraction call that includes:

- document-type assessment

- one-policy assessment
- supported fields
- candidate alternatives
- proposed page evidence

Then apply deterministic validation and evidence verification.

## **15.4 Per-page calls and aggregation**

Rejected as default due:

- cost
- duplicate candidates
- association complexity
- cross-page coverage

## **15.5 Second AI verification call**

Do not run on every document.

A second model agreeing with the first is not proof.

It may be included only in evaluation experiments.

## **15.6 Controlled structural repair**

Permit one repair attempt only when:

- provider returned content
- business meaning appears present
- failure is structural rather than factual
- the repair can use the same source and contract

## **15.7 Full extraction retry**

Use only for transient provider failures or truncated output, subject to automatic-recovery limits.

## **15.8 Accepted trade-off**

One call must perform several related extraction responsibilities, but deterministic gates prevent it from becoming sole authority.

---

## 16. Canonical extraction contract

| Canonical field           | Meaning and ownership                   | Type and cardinality           | Required for candidate             | Missing allowed                             | Evidence                  | Normalization                    | Approval           |
|---------------------------|-----------------------------------------|--------------------------------|------------------------------------|---------------------------------------------|---------------------------|----------------------------------|--------------------|
| Carrier name              | Issuing insurance carrier, policy level | Text, one                      | Yes                                | Candidate may publish with Critical warning | Page and labelled excerpt | Name spacing/case only           | Critical           |
| Policy number             | Identifier for this policy              | Text, one                      | Yes                                | Candidate may publish with Critical warning | Labelled page excerpt     | Conservative punctuation/spacing | Critical           |
| Effective date            | Policy-period start                     | Date, one                      | Yes                                | Critical warning                            | Page excerpt              | Canonical date plus source text  | Critical           |
| Expiration date           | Policy-period end                       | Date, one                      | Yes                                | Critical warning                            | Page excerpt              | Canonical date plus source text  | Critical           |
| Total policy premium      | Clearly labelled policy total           | Money, zero or one             | No                                 | Yes                                         | Clearly labelled total    | Currency and decimal form        | Import             |
| Primary named insured     | Main insured person                     | Person name, one               | Yes                                | Critical warning                            | Labelled page excerpt     | Conservative name spacing        | Critical           |
| Additional named insureds | Additional named insured people         | List of names                  | No                                 | Yes                                         | Evidence per person       | Conservative name spacing        | Import             |
| Mailing address           | Insured mailing address                 | Address, zero or one           | No                                 | Yes                                         | Labelled address block    | Structured display where safe    | Import             |
| Vehicles                  | Distinct insured vehicles               | List                           | Reviewable empty only with warning | Yes                                         | Evidence per vehicle      | Item-level                       | High-risk relation |
| Vehicle year              | Model year                              | Integer-like value per vehicle | No                                 | Yes                                         | Same row/block as vehicle | Four-digit year                  | Import             |

| Canonical field          | Meaning and ownership                              | Type and cardinality        | Required for candidate | Missing allowed | Evidence                         | Normalization                     | Approval  |
|--------------------------|----------------------------------------------------|-----------------------------|------------------------|-----------------|----------------------------------|-----------------------------------|-----------|
| Vehicle make             | Manufacturer                                       | Text per vehicle            | No                     | Yes             | Vehicle-row evidence             | Spacing/case                      | Import    |
| Vehicle model            | Model                                              | Text per vehicle            | No                     | Yes             | Vehicle-row evidence             | Spacing/case                      | Import    |
| VIN                      | Vehicle identification shown                       | Text per vehicle            | No                     | Yes             | Same vehicle row                 | Uppercase; preserve masking       | High risk |
| Comprehensive deductible | Comprehensive deductible for one vehicle           | Money per vehicle           | No                     | Yes             | Coverage and vehicle association | Currency form                     | Import    |
| Collision deductible     | Collision deductible for one vehicle               | Money per vehicle           | No                     | Yes             | Coverage and vehicle association | Currency form                     | Import    |
| Drivers                  | Distinct listed drivers                            | List of names               | No                     | Yes             | Evidence per driver              | Conservative name spacing         | Import    |
| Bodily injury limit      | Policy-level bodily injury liability               | Coverage limit, zero or one | No                     | Yes             | Labelled coverage row            | Structured split or combined form | High risk |
| Property damage limit    | Policy-level property damage liability             | Coverage limit, zero or one | No                     | Yes             | Labelled coverage row            | Canonical amount                  | High risk |
| UM/UIM limit             | Policy-level uninsured/underinsured motorist limit | Coverage limit, zero or one | No                     | Yes             | Labelled coverage row            | Preserve coverage meaning         | Import    |

## Contract rules

- Unsupported fields do not appear.
- Missing values remain missing.
- Each list item owns its own evidence.
- Vehicle deductibles must identify their vehicle.
- Liability coverage is policy-level unless the source explicitly proves another approved interpretation.
- Partial VINs remain partial.
- Source representation remains available after normalization.



---

## 17. List and relationship preservation

### 17.1 Named insureds

The model must distinguish:

- primary named insured
- additional named insured
- drivers
- agents
- household members

### 17.2 Vehicles

Each vehicle is a separate candidate item.

An item may be incomplete but must not be merged with another incomplete vehicle.

### 17.3 Drivers

Each clearly listed driver becomes one item.

Do not automatically add the primary named insured as a driver.

### 17.4 Vehicle-specific deductibles

Deductibles must attach through source evidence showing:

- vehicle identifier
- vehicle row
- vehicle number
- or an unambiguous table relationship

### 17.5 Policy-level coverage

Bodily injury and property damage limits remain policy-level.

### 17.6 Repeated vehicle rows

Rows may repeat because of:

- page continuation
- separate premium and coverage tables
- duplicate page extraction

The pipeline may consolidate only when identity evidence clearly matches.

### 17.7 Drivers on another page

Cross-page driver lists remain separate from vehicles unless the source explicitly associates them.

## 17.8 Masked VINs

Preserve visible characters and masking symbols.

## 17.9 Missing vehicle detail

Publish incomplete vehicle candidate with warnings rather than inventing the missing year, make, model, or VIN.

## 17.10 Ambiguous associations

Produce:

- Important warning when a noncritical value may belong to one of several vehicles
  - Critical warning when a deductible or identity could be attached to the wrong vehicle
- 

# 18. Field-by-field extraction semantics

## 18.1 Carrier name

### Qualifies

The company underwriting or issuing the policy.

### Common labels

- insurance company
- insurer
- company
- underwritten by
- issued by

### Do not confuse with

- agency
- producer
- broker
- servicing office
- payment processor

### Evidence

Carrier name near policy heading or issuing-company context.

### Missing

Critical.

### Conflict

Several carriers create Critical warning.

### **Normalization**

Spacing, casing, and obvious corporate-suffix formatting only.

### **OCR risks**

- dropped punctuation
- merged logo text
- agency name read as carrier

## **18.2 Policy number**

### **Qualifies**

The identifier labelled as the policy number for this declaration.

### **Common labels**

- policy number
- policy no.
- policy #
- contract number when clearly equivalent

### **Do not confuse with**

- quote number
- account number
- invoice number
- form number
- agent number
- customer ID

### **Evidence**

Label and complete source representation.

### **Missing or conflict**

Critical.

### **Normalization**

Remove only layout-introduced whitespace. Preserve meaningful hyphens and letters.

### **OCR risks**

- O/O
- I/I

- S/5
- B/8
- missing hyphens

### **18.3 Effective date**

#### **Qualifies**

Start of the policy period.

#### **Do not confuse with**

- print date
- issue date
- transaction date
- payment due date
- vehicle effective date when different

#### **Evidence**

Policy-period label and page.

#### **Missing**

Critical.

#### **Normalization**

Canonical date plus original source text.

### **18.4 Expiration date**

Same principles as effective date.

Expiration before effective date is Critical.

### **18.5 Total policy premium**

#### **Qualifies**

A clearly labelled total premium for the policy period.

#### **Common labels**

- total policy premium
- total premium
- policy premium

#### **Do not confuse with**

- installment

- amount due
- minimum payment
- vehicle premium
- coverage premium
- subtotal
- taxes or fees alone

### **Missing**

Allowed.

### **Conflict**

Several plausible totals create Important warning.

### **Normalization**

Currency and decimal representation.

### **Prohibited**

Summing components automatically.

## **18.6 Primary named insured**

### **Qualifies**

The person designated as first or primary named insured.

### **Do not confuse with**

- driver
- agent
- lienholder
- additional interest
- policy contact

### **Evidence**

Named-insured label.

### **Missing or conflict**

Critical.

## **18.7 Mailing address**

### **Qualifies**

Mailing address associated with named insured.

**Do not confuse with**

- garaging address
- agency address
- lienholder address
- carrier address
- billing-remittance address

**Missing**

Allowed with Important or Information depending on document context.

**18.8 Additional named insureds**

Extract only when clearly identified as named insureds.

Do not infer from drivers or household members.

**18.9 Vehicle year**

Must be associated with a specific vehicle.

Do not confuse with:

- policy year
- page copyright year
- model code

**18.10 Vehicle make**

Extract manufacturer, not coverage label or body style.

**18.11 Vehicle model**

Preserve model text without inventing trim.

**18.12 VIN**

Preserve exactly visible identity after conservative capitalization.

Do not complete masked or partial VINs.

**18.13 Driver name**

Extract only from a driver list or clear driver designation.

Do not duplicate the same driver because the name appears on several pages.

**18.14 Bodily injury liability limit****Qualifies**

Bodily injury liability split limit or combined limit clearly assigned to bodily injury liability.

**Do not confuse with**

- property damage
- medical payments
- UM/UIM bodily injury
- premium
- deductible

**18.15 Property damage liability limit**

Must be distinguished from:

- uninsured motorist property damage
- collision
- comprehensive
- property-damage premium

**18.16 Uninsured or underinsured motorist limit**

Extract only when the coverage label clearly identifies UM, UIM, uninsured motorist, or underinsured motorist.

Do not infer from state requirements.

**18.17 Comprehensive deductible**

Must identify:

- comprehensive or other-than-collision coverage
- deductible
- related vehicle

**18.18 Collision deductible**

Must identify:

- collision coverage
- deductible
- related vehicle

**18.19 Common OCR risk across coverages**

- currency symbols omitted
- commas removed
- split-limit slash lost
- zeroes merged
- row shifted to adjacent vehicle
- coverage abbreviations corrupted

## **19. Missing-value policy**

### **19.1 Core rule**

Absence remains absence.

### **19.2 Prohibited inference**

The pipeline must not:

- calculate premium totals
- complete VINs
- convert named insureds into drivers
- infer common coverage
- use a customer address as policy mailing address
- invent effective or expiration dates
- use general insurance knowledge
- fill a field because most policies contain it

### **19.3 State definitions**

#### **Not found**

The pipeline did not locate a supported value.

#### **Not present**

A reviewer later confirms that the document does not contain the field.

The AI pipeline cannot set this final reviewer state.

#### **Unclear**

A possible value exists but cannot be interpreted reliably.

#### **Conflicting**

Several supported candidates disagree.

#### **Unsupported field**

The document contains a value outside the Release 1 contract.

It is ignored.

#### **Extraction failure**

The pipeline could not create a structurally usable result.



## 19.4 Publication behavior

- Missing optional value: candidate may publish.
  - Missing critical identity field: publish only with Critical warning if source remains reviewable.
  - Broad extraction failure: Failed.
  - Unsupported extra data: ignore without adding a field.
- 

## 20. Value normalization

### 20.1 General principle

Preserve both:

- source representation
- normalized candidate

Normalization may improve consistency, not certainty.

### 20.2 Dates

Normalize an unambiguous source date into one canonical date representation.

Do not resolve ambiguous numeric dates without supporting U.S. document context and evidence.

### 20.3 Premium and deductibles

Normalize:

- currency symbol
- thousands separators
- decimal representation

Do not add missing decimal places when the amount is ambiguous.

### 20.4 Policy numbers

Allowed:

- trim surrounding whitespace
- remove line-wrap spaces
- normalize casing when safe

Prohibited:

- changing O to 0
- inserting missing characters
- removing meaningful punctuation without evidence

## 20.5 Names

Allowed:

- whitespace cleanup
- safe casing for display

Prohibited:

- expanding initials
- switching name order
- guessing spelling

## 20.6 Addresses

Allowed:

- whitespace and line joining
- conservative punctuation cleanup

Prohibited:

- geocoding-based replacement
- postal standardization that changes uncertain content
- filling missing city, state, or ZIP

## 20.7 VIN

- uppercase letters
- preserve masking
- preserve source characters
- flag invalid-looking complete VIN
- never repair OCR confusion automatically

## 20.8 Vehicle year

Normalize only clear four-digit years.

## 20.9 Make and model

Whitespace and casing only.

## 20.10 Liability split limits

Preserve whether the source shows:

- per-person/per-accident
- combined single limit
- one amount
- abbreviated thousands

Do not transform a split limit into a combined limit.

## 20.11 Common OCR confusion

OCR substitutions may generate candidate alternatives and warnings, but normalization must not silently choose a corrected character.

---

## 21. Source-evidence contract

### 21.1 Required evidence properties

For each extracted value:

- source page
- supporting text excerpt
- evidence source type
- source representation
- exact location when reliable
- additional page when required
- relationship context for lists

### 21.2 Critical fields

Carrier, policy number, dates, and primary named insured require:

- at least one supporting page
- a labelled or strongly contextual excerpt
- deterministic confirmation that the excerpt exists in source artifacts

### 21.3 Vehicles

Evidence should show enough of the same row or block to connect:

- year
- make
- model
- VIN

### 21.4 Drivers

Each driver requires evidence from a driver list or designation.

### 21.5 Policy-level coverage

Evidence should include:

- coverage label
- limit
- table context where useful

## 21.6 Vehicle deductibles

Evidence must show the deductible's relationship to the relevant vehicle.

## 21.7 Exact location unavailable

Use:

- page
- supporting excerpt
- "Exact location unavailable"

Do not invent coordinates.

## 21.8 Poor OCR evidence

Show:

- OCR source
- uncertain excerpt
- page
- OCR Concern warning

## 21.9 Several candidates

Retain evidence for each candidate.

## 21.10 Manual search

When exact evidence cannot be found but the value may still be valid:

- create Low Evidence warning
- provide page-only evidence
- require reviewer inspection

---

## 22. Evidence-generation strategy

### 22.1 Model-only evidence

#### Risk

A model may provide an excerpt or location that does not exist.

Insufficient by itself.

### 22.2 Deterministic source search

#### Strength

Proves the excerpt exists.

### **Limitation**

Normalized values may not exactly match OCR or source representation.

## **22.3 Combined strategy**

### **Recommendation**

1. Ask the model for:
2. source page
3. source representation
4. short supporting excerpt
5. Verify:
6. page exists
7. excerpt or close source representation exists
8. label and value relationship are plausible
9. Link positional data from direct text or OCR when available.
10. downgrade evidence when verification fails.

## **22.4 OCR bounding boxes**

Use OCR coordinates when:

- provider or local OCR provides them
- text alignment is reliable
- page transform is known

## **22.5 Page-only evidence**

Permitted when:

- exact text location cannot be verified
- page is still reviewable
- field receives Low Evidence warning

## **22.6 Fabricated evidence prevention**

Never publish a model-supplied excerpt as verified until matched to source artifacts.

## **22.7 Normalized-to-source mapping**

Retain model-supplied source representation and search using:

- exact text

- whitespace-normalized text
- safe punctuation normalization
- known source formatting

Do not use broad fuzzy matching that could attach the wrong amount.

## **22.8 Correct value without exact match**

Publish with:

- page evidence
- unverified excerpt state
- Important Low Evidence warning

when a reviewer can inspect it safely.

---

## **23. AI extraction instruction specification**

The extraction instruction should contain the following conceptual sections.

### **23.1 Role and scope**

- read one supported personal auto declaration
- produce candidate information only
- do not make insurance decisions

### **23.2 Allowed fields**

List only the approved canonical fields.

### **23.3 Source grounding**

- use only supplied document content
- identify supporting page
- preserve source representation
- report competing candidates

### **23.4 Missing-value rule**

- return missing when unsupported
- do not guess
- do not calculate absent values

### **23.5 Document classification**

- supported
- unsupported
- ambiguous
- several-policy risk

## 23.6 Relationship rules

- keep vehicles separate
- keep drivers separate
- keep named insureds separate
- attach deductibles to vehicles
- keep liability coverage at policy level

## 23.7 OCR caution

- OCR text may contain errors
- preserve suspicious source representation
- do not automatically repair identifier characters

## 23.8 Evidence behavior

- page
- source excerpt
- source value
- alternatives

## 23.9 Prohibited behavior

- recommendations
- coverage adequacy
- policy verification
- unsupported fields
- assumptions from insurance knowledge

## 23.10 Contract adherence

The model must follow the canonical extraction contract.

## 23.11 Versioning

Every instruction change receives a new instruction version.

Material wording changes require regression testing.

---

# 24. Structured-output validation

## 24.1 Failure conditions

- required structural group missing
- wrong value type
- unexpected top-level field
- malformed list
- duplicate object
- value outside contract
- broken vehicle-deductible relationship
- evidence missing

- empty result with supported source
- provider refusal
- truncated output

## 24.2 Immediate deterministic repair

Allowed for purely mechanical issues such as:

- harmless empty optional section
- duplicated whitespace
- known provider wrapper
- safe conversion of an explicitly numeric year

It must not alter business meaning.

## 24.3 Controlled model repair

Allow one repair attempt when:

- output contains the needed information
- failure is structural
- original source context and contract remain available
- no factual reinterpretation is requested

## 24.4 Complete retry

Use for:

- truncated response
- transient provider failure
- empty provider output
- refusal caused by temporary service behavior

## 24.5 Recommendation

Use:

1. strict structural validation
2. deterministic safe repair
3. one controlled model-based structural repair
4. final failure

## 24.6 Prohibited loops

No unlimited:

- repair
- retry
- “ask again until valid”
- alternate-model sequence



## 25. Deterministic field validation

| Validation                                   | Candidate outcome                                            |
|----------------------------------------------|--------------------------------------------------------------|
| Carrier missing                              | Needs Review with Critical warning when source is reviewable |
| Policy number missing                        | Needs Review with Critical warning                           |
| Several policy numbers                       | Critical warning or Failed when several policies are clear   |
| Effective date missing                       | Critical warning                                             |
| Expiration date missing                      | Critical warning                                             |
| Expiration before effective date             | Critical warning                                             |
| Premium malformed                            | Important warning or missing                                 |
| Complete VIN invalid-looking                 | Important or Critical based on association                   |
| Partial/masked VIN                           | Important warning; preserve partial form                     |
| Vehicle year implausible                     | Important warning                                            |
| Duplicate vehicles                           | Important or Critical warning                                |
| Duplicate drivers                            | Important warning                                            |
| Address incomplete                           | Important warning                                            |
| Liability structure malformed                | Important warning                                            |
| Deductible not attached to vehicle           | Critical association warning                                 |
| Customer-name conflict                       | Critical warning                                             |
| Customer-address conflict                    | Important warning                                            |
| Supported candidate has no critical evidence | Critical or Low Evidence warning                             |
| Validation configuration fails               | Failed; do not publish candidate                             |

### 25.1 Candidate-publication blocking

Block publication when:

- structural result cannot be normalized
- source pages cannot be identified
- validation engine cannot execute
- document is unreadable
- no meaningful supported content exists
- internal contract versions are incompatible

## 25.2 Human-review conditions

Most factual conflicts should publish as warnings when:

- source remains inspectable
  - candidate alternatives are preserved
  - reviewer can resolve the issue
- 

## 26. Uncertainty and warning generation

### 26.1 Information

Signals include:

- OCR used without field-specific concern
- optional field missing
- formatting normalized
- exact location unavailable but page evidence strong
- stored result used

### 26.2 Important

Signals include:

- weak evidence
- partial VIN
- ambiguous optional premium
- incomplete vehicle detail
- address conflict
- duplicate driver
- low OCR quality affecting noncritical field
- optional coverage conflict

### 26.3 Critical

Signals include:

- conflicting policy numbers
- missing critical identity
- invalid policy period
- wrong-customer risk
- several policies
- vehicle-deductible association ambiguity
- unsupported document ambiguity
- no reliable evidence for critical value

## 26.4 Signal sources

Warnings should use:

- deterministic validation
- evidence verification
- candidate alternatives
- text quality
- OCR quality
- document classification
- customer comparison
- list association

## 26.5 Model confidence

A model-supplied confidence value may be retained diagnostically during evaluation but cannot determine warning severity.

## 26.6 Combining several signals

For one field:

- display the highest severity
- combine related reasons into one explanation
- preserve individual signals in detail
- avoid producing several nearly identical warnings

Example:

Instead of three separate warnings for an OCR-derived incomplete VIN, show:

Important: VIN appears incomplete and came from a low-quality scanned row. Check the source characters.

---

# 27. Candidate publication gate

## 27.1 Required conditions

A result may become reviewable only when:

- source PDF passed safety inspection
- enough pages are readable
- document is supported or reviewably ambiguous
- several-policy state is resolved or explicitly Critical
- structured result is valid
- candidate follows the canonical contract
- normalization completed
- validation completed
- warning generation completed
- critical fields have evidence or explicit evidence warnings

- candidate and pipeline versions are recorded
- no fatal stage error remains

## **27.2 Needs Review**

Use when:

- result is structurally valid
- reviewer can inspect source
- conflicts are explicit
- missing fields are explicit
- warnings identify unsafe values

## **27.3 Failed**

Use when:

- processing cannot produce a safe reviewable candidate
- source cannot be read
- OCR fails broadly
- provider output remains invalid
- evidence preparation fails globally
- pipeline versions are incompatible

## **27.4 Unsupported Document**

This is a Failed category when the file clearly falls outside Release 1.

## **27.5 Needs Review with Critical warnings**

Use for reviewable cases such as:

- one of two policy-number candidates must be selected
- customer identity differs
- policy dates conflict
- one-policy status is ambiguous but likely resolvable

## **27.6 Important meaning**

Passing this gate means:

Safe enough for human inspection.

It does not mean correct, approved, or verified.

---

## 28. Failure taxonomy

| Failure                      | Stage          | Retryable?        | Replacement? | Paid use possible? | Preserved artifacts                    | User category                   |
|------------------------------|----------------|-------------------|--------------|--------------------|----------------------------------------|---------------------------------|
| Not a PDF                    | Preflight      | No                | Yes          | No                 | File identity only                     | Unsupported file                |
| Malformed PDF                | Inspection     | Usually no        | Yes          | No                 | Inspection result                      | Damaged PDF                     |
| Password protected           | Inspection     | No                | Yes          | No                 | Encryption result                      | Locked PDF                      |
| No readable pages            | Text/OCR       | Sometimes         | Usually      | Possibly           | Inspection and OCR diagnostics         | No readable pages               |
| Direct-text failure          | Parser         | Sometimes         | No           | No                 | Source and parser error                | Document text could not be read |
| OCR failure                  | OCR            | Sometimes         | Possibly     | Yes                | Direct text, OCR request metadata      | Scanned pages could not be read |
| Unsupported document         | Classification | No                | Yes          | Maybe              | Text and classification evidence       | Unsupported document            |
| Several inseparable policies | Policy check   | No                | Yes          | Maybe              | Policy candidates                      | Several policies in one file    |
| Context preparation failure  | Preparation    | Sometimes         | No           | No                 | Source artifacts                       | Document could not be prepared  |
| Provider timeout             | AI             | Yes               | No           | Yes                | Prepared context and request reference | Processing service timed out    |
| Provider unavailable         | AI/OCR         | Yes               | No           | Possibly           | Stage artifacts                        | Processing service unavailable  |
| Provider refusal             | AI             | Sometimes         | No           | Yes                | Refusal category                       | Result could not be created     |
| Invalid structured output    | Structure      | Yes within limits | No           | Yes                | Raw response for limited retention     | Invalid extraction result       |

| Failure                | Stage             | Retryable?                | Replacement? | Paid use possible? | Preserved artifacts        | User category                  |
|------------------------|-------------------|---------------------------|--------------|--------------------|----------------------------|--------------------------------|
| Repair failure         | Structure         | Yes through user retry    | No           | Yes                | Original and repair result | Invalid extraction result      |
| Evidence failure       | Evidence          | Sometimes                 | No           | No                 | Candidate and source       | Evidence could not be prepared |
| Validation failure     | Validation engine | Yes after fix             | No           | No                 | Candidate                  | Internal processing error      |
| Duration exceeded      | Any stage         | Yes                       | No           | Possibly           | Completed stages           | Processing took too long       |
| Configuration mismatch | Publication       | No until deployment fixed | No           | No                 | All artifacts              | Processing configuration error |

### Audit expectation

Audit should record user-relevant failure outcomes. Technical details and provider traces remain in diagnostic logs.

## 29. Internal recovery, Technical Retry, and Reprocessing

### 29.1 Automatic internal recovery

Permitted for narrowly transient events:

- temporary provider timeout
- temporary network failure
- worker interruption before stage completion

It must be limited and recorded diagnostically.

### 29.2 Technical Retry

Creates a new attempt after final technical failure.

May reuse artifacts when valid.

### 29.3 Intentional Reprocessing

Creates a new attempt because:

- model changed
- instructions changed

- parser changed
- OCR changed
- result quality is being re-evaluated

## **29.4 Stored Result replay**

Uses approved stored artifacts without external processing.

## **29.5 Replacement upload**

Required when the source file changes.

## **29.6 Stage reuse**

### **May reuse**

- source PDF
- safety inspection
- direct text
- page render
- OCR result

when their producing configuration and source identity remain valid.

### **Must rerun**

- AI extraction after model or instruction change
- normalization after normalizer change
- validation after validation change
- evidence after evidence logic change

## **29.7 OCR reuse**

Do not repeat OCR when:

- source is identical
- OCR engine and preprocessing configuration are unchanged
- retained output is complete

## **29.8 Approved policy protection**

No retry or reprocessing can change an approved version automatically.

## **29.9 Retry enforcement**

The Processing module counts user-triggered Technical Retries and rejects a third.

## **29.10 Reproducibility**

Each attempt identifies reused and newly generated stages.

## 30. Stage-level caching and artifact reuse

| Artifact       | Reusable when                                               | Invalidated by                                        |
|----------------|-------------------------------------------------------------|-------------------------------------------------------|
| PDF inspection | Same source bytes and inspection configuration              | Source or safety-rule change                          |
| Direct text    | Same source and parser version/configuration                | Parser or extraction settings change                  |
| OCR text       | Same rendered pages and OCR configuration                   | OCR engine, preprocessing, orientation, render change |
| Page images    | Same source, page transform and render configuration        | Render settings or source change                      |
| Classification | Same selected text/image context and classifier rules       | Context, rules, model or instruction change           |
| AI candidate   | Same complete pipeline input and model/instruction contract | Model, instruction, contract or context change        |
| Normalization  | Same candidate and normalizer                               | Normalizer or contract change                         |
| Validation     | Same normalized candidate and validation rules              | Validator or customer-context change                  |
| Evidence       | Same source artifacts, candidate and evidence generator     | Parser/OCR/evidence/candidate change                  |
| Stored Result  | Same candidate contract and compatible workflow version     | Contract or evidence incompatibility                  |

### 30.1 Cache identity

Reuse must be based on source and configuration identity, not file name or approximate similarity.

### 30.2 Stale-artifact warning

A reused artifact must never be shown as newly generated live output.

### 30.3 Cost control

Stage reuse should allow:

- frontend development without providers
- prompt evaluation without repeated OCR
- validation testing without repeated model calls
- evidence changes without rerunning OCR where possible



## 31. Benchmark-corpus design

### 31.1 Initial development corpus

**Recommended size:** 12 documents

Include:

- 4 clean text PDFs
- 3 readable scans
- 1 mixed text/scan PDF
- 1 rotated scan
- 1 multiple-vehicle case
- 1 missing-optional-fields case
- 1 unsupported document

Purpose:

- develop the first contract
- validate parser output
- establish pipeline mechanics

### 31.2 Provider-comparison corpus

**Recommended size:** 24–30 documents

Include:

- at least 8 clean layouts
- at least 8 scanned or mixed documents
- multi-vehicle and multi-driver cases
- masked VINs
- coverage tables
- ambiguous premiums
- several-policy and unsupported examples

Purpose:

- compare OCR and model configurations

### 31.3 Fixed release-gate corpus

**Recommended size:** 40–50 documents

Balanced across:

- clean
- scanned
- mixed
- rotated
- table-heavy

- list-heavy
- missing
- conflicting
- unsupported
- failure conditions

### 31.4 Hidden holdout

**Recommended size:** 10–15 documents

No prompt or threshold tuning based on these documents.

### 31.5 Failure corpus

**Recommended size:** 10–15 artifacts

Include:

- corrupt PDF
- locked PDF
- zero-page or malformed test
- unreadable scan
- provider timeout simulation
- invalid structure fixture
- unsupported document
- several-policy file

### 31.6 Public-demo sample set

**Recommended size:** 5–6 synthetic documents

- clean success
- scan with corrections
- warning-heavy case
- failed unsupported case
- approved replacement case
- historical policy case

### 31.7 Credibility without research-scale data

Report both:

- percentages
- raw counts

A benchmark of 40 documents does not prove universal carrier support. It proves performance on a documented Release 1 corpus.

## 32. Benchmark-document sourcing and privacy

### 32.1 Acceptable sources

- fully synthetic documents
- self-created carrier-like layouts
- documents the owner has explicit rights to process
- client samples with explicit permission
- carefully controlled private evaluation files
- redacted copies when redaction does not destroy the evaluation feature

### 32.2 Unacceptable sources

- random declarations found online
- leaked or uncontrolled personal documents
- documents copied into public demos without permission
- files with unclear distribution rights
- private documents uploaded to several providers without approval

### 32.3 Synthetic realism

Synthetic documents should vary:

- carrier-like branding
- table structure
- page order
- vehicle count
- driver count
- policy-number patterns
- coverage formats
- scan quality
- address format

They must not all derive from one visual template.

### 32.4 Public versus private sets

#### Public demo

Synthetic only.

#### Private evaluation

May contain permitted real-world complexity under restricted access.

### 32.5 Redaction limits

Redaction can alter:

- OCR quality
- spacing

- layout
- identifier patterns
- page images

A redacted document may not be equivalent to the original extraction problem.

## 32.6 Storage

Private evaluation files require:

- private storage
  - restricted access
  - limited provider transmission
  - explicit retention
  - deletion procedure
- 

## 33. Ground-truth creation

### 33.1 Solo-developer process

1. Read the complete document manually.
2. Annotate every supported field.
3. Mark source page.
4. capture supporting excerpt.
5. record source representation.
6. identify relationships.
7. mark fields absent, ambiguous, or conflicting.
8. review critical fields a second time on another day or separate pass.
9. freeze a ground-truth version.

### 33.2 Critical second review

At minimum, recheck:

- carrier
- policy number
- dates
- primary named insured
- VINs
- liability limits
- vehicle deductibles

### 33.3 Ground-truth states

Each field can be:

- present and unambiguous
- absent
- ambiguous
- conflicting

- unreadable
- not applicable

### **33.4 Evidence truth**

Ground truth includes:

- correct page
- correct excerpt
- exact location when feasible
- relationship context

### **33.5 Unsupported-document truth**

Record:

- actual document type
- reason unsupported
- whether a declaration page could be confused with it

### **33.6 Annotation correction**

Ground truth is versioned. When an annotation error is found:

- correct it
- record the correction
- rerun affected evaluation
- do not count the old expected value as a model failure

### **33.7 Why annotation quality matters**

A model cannot be evaluated accurately against incorrect or incomplete ground truth.

---

## **34. Development, tuning, and holdout separation**

### **34.1 Development set**

Used freely for:

- parser work
- context design
- initial prompts
- normalizer
- warnings

### **34.2 Prompt-tuning set**

Used to adjust:

- extraction instruction

- model context
- candidate alternatives
- evidence behavior

### **34.3 Provider-comparison set**

Used with frozen instructions and contract to compare providers fairly.

### **34.4 Hidden holdout**

Used only after provisional configuration selection.

### **34.5 Regression set**

Fixed documents representing previously solved errors.

### **34.6 Public-demo set**

Optimized for demonstration coverage, not provider selection.

### **34.7 Overfitting risks**

#### **Prompt overfitting**

Instructions mention specific sample wording or layout.

#### **Layout overfitting**

Pipeline works on one synthetic template.

#### **Carrier-like overfitting**

Rules depend on one brand pattern.

### **34.8 New failure samples**

When a new permitted document exposes a failure:

1. diagnose it
2. create or obtain an approved regression example
3. add it after ground-truth review
4. avoid tuning solely for one file without checking the wider corpus

---

## **35. Evaluation metrics**

### **35.1 Document classification**

How often supported and unsupported documents receive the correct class.

Report false supported and false unsupported separately.

### **35.2 One-policy detection**

How often the system correctly identifies:

- one policy
- reviewable ambiguity
- several policies

### **35.3 Field extraction correctness**

Percentage of expected field values extracted correctly.

Report per field and field group.

### **35.4 Normalized-value correctness**

Whether the canonical value preserves the source meaning.

### **35.5 List-item completeness**

Whether all expected vehicles, drivers, and named insureds were included.

### **35.6 Vehicle association accuracy**

Whether each vehicle's details and deductibles remain attached to the correct vehicle.

### **35.7 Driver separation**

Whether drivers are neither merged nor duplicated.

### **35.8 Coverage association**

Whether policy-level limits remain policy-level and vehicle deductibles stay vehicle-specific.

### **35.9 Evidence-page correctness**

Whether the cited page actually supports the value.

### **35.10 Evidence-excerpt support**

Whether the excerpt supports the extracted meaning.

### **35.11 Exact-location accuracy**

When coordinates are provided, whether they identify the correct source region.

### **35.12 Missing-value behavior**

Measure:

- false invention
- correct missing
- incorrect missing
- conflict correctly preserved

### **35.13 Unsupported-field leakage**

Count fields outside the approved contract that appear in the candidate.

Target: zero.

### **35.14 Critical silent errors**

Wrong critical values published without an appropriate blocking or attention warning.

### **35.15 Warning precision**

Of generated warnings, how many identify a real review issue?

### **35.16 Warning recall**

Of real issues, how many receive an appropriate warning?

### **35.17 Structural success**

How often the result passes structural validation, including after one allowed repair.

### **35.18 Repeatability**

Stability of values, lists, evidence, and warnings across repeated runs.

### **35.19 Latency**

Time per stage and total processing time.

### **35.20 Usage**

Pages, image input, text input, output, OCR operations, repair calls.

### **35.21 Cost**

Estimated or provider-reported cost per document and benchmark run.

### **35.22 Technical failure rate**

Percentage of otherwise valid documents that fail due to infrastructure or provider behavior.



## 36. Criticality and weighted evaluation

### 36.1 Critical identity fields

- carrier
- policy number
- effective date
- expiration date
- primary named insured

These receive the strictest reporting.

### 36.2 High-risk operational fields

Report separately:

- VIN
- bodily injury limits
- property damage limits
- vehicle-deductible associations

A wrong VIN or wrongly associated deductible can be operationally serious even though Release 1 allows the value to be missing.

### 36.3 Important fields

- mailing address
- vehicle year, make and model
- drivers
- additional named insureds

### 36.4 Optional fields

- total premium when not clearly present
- UM/UIM
- comprehensive deductible
- collision deductible

Optional does not mean a wrong extracted value is harmless.

### 36.5 List and relationship reporting

Use separate metrics for:

- item presence
- item duplication
- field-to-item association
- missing list
- wrong list membership

## 36.6 Silent critical error definition

A candidate value for a critical identity field is wrong, but the pipeline publishes it without a Critical or appropriate Important warning that directs the reviewer to inspect that value.

## 36.7 Weighted score

A weighted score may summarize configurations, but it must never replace:

- raw critical-field results
  - silent-error count
  - relationship results
  - evidence results
- 

# 37. Initial quality targets and release gates

## 37.1 Clean text PDFs

- overall supported-field correctness: **at least 95%**
- critical identity correctness: **at least 98%**
- structural success after at most one repair: **at least 99%**
- unsupported-field leakage: **zero**

## 37.2 Readable scanned PDFs

- overall supported-field correctness: **at least 90%**
- critical identity correctness: **at least 95%**
- OCR technical success on readable scans: **at least 95%**

## 37.3 Silent critical errors

**Zero** on the release-gate and hidden holdout sets.

This is a zero-tolerance rule because an average can hide a serious wrong value.

## 37.4 Supported-document classification

- supported-document recall: provisional target **at least 95%**
- unsupported documents silently accepted without warning: **zero**
- clearly unrelated documents sent through full extraction: minimize and report

## 37.5 One-policy detection

- clearly separate policies silently merged: **zero**
- reviewable ambiguous multi-policy cases correctly warned: **all benchmark cases**

## 37.6 Lists

- clean list-item completeness: **at least 95%**
- scanned list-item completeness: **at least 90%**

- unflagged wrong vehicle association: **zero**
- unflagged driver merge: **zero**

### 37.7 Evidence

- critical-field correct page: **at least 98%**
- critical-field excerpt support: **at least 95%**
- fabricated excerpt accepted as verified: **zero**
- wrong exact bounding box presented as verified: **zero on evaluated boxes**

### 37.8 Warnings

- all silent critical-error scenarios must instead generate an appropriate warning
- Important-warning precision should be high enough that reviewers do not face systematic warning noise
- report warning precision and recall by category rather than one combined number

A provisional target of **at least 80% precision and 90% recall** for important review issues should be validated against corpus size.

### 37.9 Repeatability

Across repeated runs:

- critical field values stable in all runs, or every variation explicitly warned
- list membership stable in at least 95% of repeated runs
- structure succeeds every run
- evidence pages stable for critical fields

### 37.10 Latency

Hypotheses to validate:

- clean PDFs: median near or below one minute
- OCR PDFs: median near or below two minutes
- slower-tail target defined separately

### 37.11 Cost

A production configuration must remain below the owner-approved:

- cost per clean document
- cost per scanned document
- full benchmark-run budget
- public-demo monthly live allowance

### 37.12 Technical failure

Excluding intentionally malformed documents:

- clean valid-PDF failure rate below 5%
- readable-scan technical failure rate below 10%

- no unrecoverable failure caused by normal browser closure

### 37.13 Small-corpus reporting

Every percentage must include numerator and denominator.

These are internal gates, not marketing claims.

---

## 38. Repeatability testing

### 38.1 Recommended run count

Run **five independent live extractions** for selected documents.

### 38.2 Document selection

Use 8–10 documents covering:

- clean text
- scan
- multiple vehicles
- multiple drivers
- coverage table
- masked VIN
- ambiguous premium
- critical conflict

### 38.3 Measure

- critical field values
- noncritical values
- item count
- list ordering
- associations
- evidence pages
- evidence excerpts
- warnings
- structural validity
- latency
- usage
- estimated cost

### 38.4 Harmless variation

May include:

- harmless whitespace
- equivalent casing
- evidence excerpt containing slightly different surrounding text
- list ordering that preserves all associations, although stable order remains preferable

## 38.5 Blocking variation

- policy number changes
- date changes
- named insured changes
- vehicle merge or split
- deductible changes vehicle
- coverage becomes missing or invented
- evidence page changes incorrectly
- warning severity drops on a risky value
- structural failure in any run

## 38.6 Model determinism

Do not claim deterministic output. The pipeline should reduce variation through:

- fixed instructions
  - fixed contract
  - stable context
  - provider controls where available
  - deterministic normalization
  - deterministic validation
- 

# 39. Latency evaluation

## 39.1 Measure separately

- queue waiting
- storage access
- safety inspection
- direct extraction
- image rendering
- OCR
- context preparation
- AI call
- structural repair
- normalization
- validation
- evidence
- publication

## 39.2 Report

- median
- 90th percentile or slower-tail measurement
- minimum and maximum
- warm versus cold infrastructure
- provider timeout count

### **39.3 Clean-document hypothesis**

Approximately one minute is a product target, not a promise.

### **39.4 Scan hypothesis**

Approximately two minutes is a product target, not a promise.

### **39.5 Cold starts**

Report separately because free or sleeping infrastructure can dominate processing time.

### **39.6 Stored Results**

Replay latency is a demonstration setting and must not be mixed with live-pipeline latency.

### **39.7 Delayed behavior**

Define the user-visible delayed threshold based on observed slower-tail latency rather than guessing before evaluation.

---

## **40. Cost evaluation**

### **40.1 Cost components**

#### **Direct extraction**

Normally local compute with no per-document provider charge.

#### **OCR**

- local worker compute
- or managed per-page charge

#### **AI input**

Text and image input usage.

#### **AI output**

Structured result and evidence output.

#### **Repairs**

Additional provider call.

#### **Retries and reprocessing**

Separate attempts.

## **Failure**

A failed provider call may still incur usage.

## **Stored Results**

No live OCR or AI provider usage.

## **40.2 Current model list pricing snapshot**

As of August 1, 2026:

- OpenAI GPT-5.6 Luna lists \$0.20 per million input tokens and \$1.20 per million output tokens.
- OpenAI GPT-5.6 Terra lists \$2 per million input tokens and \$12 per million output tokens.
- Claude Sonnet 5 lists \$3 per million input tokens and \$15 per million output tokens, with a temporary introductory price documented through August 31, 2026.
- Gemini 3.6 Flash lists \$1.50 per million input tokens and \$7.50 per million output tokens.
- Gemini 3.5 Flash-Lite lists \$0.30 per million input tokens and \$2.50 per million output tokens.

These are official list prices at the research date and may change.

## **40.3 Cost per document**

Calculate from measured:

- input text tokens
- image or PDF input
- output tokens
- OCR pages
- repair attempts
- provider-specific pricing

Do not estimate from file size alone.

## **40.4 Required reports**

- average clean-document cost
- slower-tail clean cost
- average scanned-document cost
- failed-document cost
- full benchmark-run cost
- five-run repeatability cost
- public-demo allowance consumption

## **40.5 Maximum acceptable cost**

The approved documents do not set an exact monetary ceiling.

The owner must approve:

- target average clean-document cost
- target average scanned-document cost

- total evaluation-spike budget
- monthly public-demo live-processing budget

## 40.6 No invented precision

Where provider billing details are unavailable:

- report tokens
- pages
- calls
- runtime
- state that cost remains estimated

---

## 41. AI model/provider evaluation

### 41.1 Current capability facts

#### OpenAI

Current GPT-5.6 models support text and image input, structured outputs, and large context windows. OpenAI's PDF input flow can supply both PDF text and page images to vision-capable models. Structured Outputs enforce schema adherence rather than only valid JSON. OpenAI states that API data is not used for model training by default; standard abuse-monitoring logs may be retained for up to 30 days unless different approved controls apply.

#### Anthropic

Current Claude models include Claude Sonnet 5, Opus 5, Fable 5, and Haiku 4.5. Claude's API supports PDF input, including text and visual content, and structured outputs on current supported model families. Anthropic documents that structured outputs and citations have compatibility limitations when used together, which supports using product-owned evidence verification rather than depending on provider citations. Retention and Zero Data Retention eligibility depend on the model and commercial arrangement.

#### Google

Current stable Gemini models include Gemini 3.6 Flash and Gemini 3.5 Flash families. Gemini supports native document understanding and structured outputs based on JSON Schema concepts. Google documents that uploaded Files API content is retained temporarily and that paid-service prompts and responses are not used to improve products under its paid-service terms, subject to current abuse-monitoring and service conditions.

### 41.2 Shortlist

#### Candidate A: Gemini 3.6 Flash

##### Expected fit

- document and image support
- structured outputs



- competitive multimodal capability
- suitable for text-plus-selected-image strategy
- moderate current list cost

#### **Risks**

- native document capabilities may tempt the pipeline to skip reproducible parsing
- quality on insurance associations is unproven
- provider file-retention behavior must match selected API flow

#### **Candidate B: OpenAI GPT-5.6 Luna**

##### **Expected fit**

- very low current listed token cost
- structured outputs
- text and image input
- strong cost benchmark

#### **Risks**

- low price does not prove document accuracy
- may require stronger model tier for complex tables
- PDF visual input may increase token use

#### **Reserve Candidate C: Claude Sonnet 5**

##### **Expected fit**

- large context
- structured output
- PDF and visual reasoning

#### **Risks**

- higher listed cost
- citations and structured-output constraints complicate provider-native evidence
- quality advantage is unproven

### **41.3 Provisional primary choice**

**Gemini 3.6 Flash** is the provisional primary benchmark configuration because it provides a balanced current combination of document input, multimodal capability, structured output, and moderate pricing.

### **41.4 Second benchmark candidate**

**OpenAI GPT-5.6 Luna** is the cost-focused challenger.

## 41.5 Claude benchmark role

Run Claude Sonnet 5 on a smaller difficult-document subset when:

- Gemini and Luna both miss association targets
- complex coverage tables remain weak
- a quality ceiling needs investigation

## 41.6 Benchmark reversal

Choose GPT-5.6 Luna if it meets all quality and repeatability gates at materially lower cost.

Choose Claude Sonnet 5 only if it provides a material quality advantage on critical fields, evidence, and associations that justifies its cost.

## 41.7 Rate limits

Provider rate limits depend on account and service tier. They should be recorded during the evaluation spike rather than assumed from general product documentation.

## 41.8 Production rule

Only the winning model remains enabled in Release 1 production.

---

## 42. OCR-provider evaluation

| Criterion              | OCRmyPDF/Tesseract                   | Google Enterprise Document OCR                        |
|------------------------|--------------------------------------|-------------------------------------------------------|
| Per-page provider cost | None; worker compute only            | Current managed page pricing                          |
| Hosting burden         | Higher                               | Lower                                                 |
| Privacy                | Local processing                     | External transmission                                 |
| Bounding boxes         | Available through OCR output         | Managed layout coordinates                            |
| Rotation               | Supported through OCR workflow       | Managed orientation capabilities                      |
| Table handling         | Basic; depends on reading order      | Stronger expected layout handling, unproven on corpus |
| Repeatability          | High for fixed binaries/settings     | Provider behavior may evolve                          |
| Latency                | Depends on worker CPU                | Network plus managed processing                       |
| Deployment burden      | Native packages and worker resources | Credentials and API integration                       |
| Evidence quality       | Good when OCR boxes align            | Potentially strong                                    |

| Criterion       | OCRmyPDF/Tesseract | Google Enterprise Document OCR |
|-----------------|--------------------|--------------------------------|
| Lock-in         | Low                | Medium                         |
| Development use | Excellent          | Intentional live tests         |

## 42.1 Provisional primary

OCRmyPDF/Tesseract.

## 42.2 Conditional managed alternative

Google Enterprise Document OCR.

## 42.3 Benchmark gates

Managed OCR should replace local OCR when it materially improves:

- critical OCR field accuracy
- VIN accuracy
- policy-number accuracy
- vehicle-row preservation
- evidence coordinates
- rotated-page success
- technical reliability

## 42.4 Resource escalation rule

Managed OCR also wins if local OCR meets accuracy targets but exceeds acceptable:

- worker memory
- CPU time
- total latency
- deployment complexity

## 42.5 Production simplicity

Do not dynamically choose local versus managed OCR per document in Release 1 unless the benchmark proves a clear deterministic rule and the added maintenance is justified.

# 43. Provisional Release 1 provider recommendation

| Capability    | Provisional recommendation | Why                                       | Main alternative | Trade-off           | Benchmark reversal                                   |
|---------------|----------------------------|-------------------------------------------|------------------|---------------------|------------------------------------------------------|
| Direct parser | PyMuPDF                    | Text, positions and rendering in one tool | pypdf            | Stronger dependency | pypdf materially outperforms or licensing blocks use |

| Capability             | Provisional recommendation                        | Why                                            | Main alternative           | Trade-off                                  | Benchmark reversal                                                    |
|------------------------|---------------------------------------------------|------------------------------------------------|----------------------------|--------------------------------------------|-----------------------------------------------------------------------|
| OCR                    | OCRmyPDF/<br>Tesseract                            | Low cost, local, repeatable                    | Google Enterprise OCR      | Worker resource use                        | Managed OCR materially improves critical scan accuracy or reliability |
| AI model               | Gemini 3.6 Flash                                  | Balanced document, image and structure support | GPT-5.6 Luna               | Higher list cost than Luna                 | Luna passes all gates at lower cost                                   |
| Difficult-case reserve | Claude Sonnet 5                                   | Quality ceiling comparison                     | Higher OpenAI model tier   | Higher cost                                | Use only if material quality gain                                     |
| Input                  | Page-aware text plus selected images              | Preserves evidence and controls cost           | Native full PDF            | More context-preparation work              | Full PDF wins quality, repeatability and cost gates                   |
| Calls                  | One extraction call plus deterministic validation | Lowest reasonable complexity                   | Separate verification call | One model call carries more responsibility | Add second call only if benchmark proves required                     |
| Structure              | Provider structured output plus local validation  | Strong contract adherence                      | Free-form parsing          | Provider schema constraints                | Reconsider if provider structure fails repeatedly                     |
| Evidence               | Model proposal plus deterministic verification    | Prevents fabricated evidence                   | Provider-only citations    | More evidence code                         | Page-only evidence if location matching proves unreliable             |
| Raw retention          | Short configurable diagnostic retention           | Limits PII accumulation                        | Keep indefinitely          | Less long-term diagnosis                   | Extend only for approved benchmark samples                            |

All quality conclusions above are provisional until the evaluation spike.

## 44. Evaluation execution plan

### Step 1: Prepare benchmark documents

Create and classify the initial corpus.

### Step 2: Complete ground truth

Manually annotate supported fields, evidence, relationships, and ambiguity.

### Step 3: Validate direct extraction

Compare PyMuPDF and pypdf on:

- extraction success
- page text
- positions
- reading order
- rendering
- resource use

Eliminate a parser if it fails foundational requirements.

### Step 4: Compare OCR

Run local and managed OCR on the same scan subset.

Do not tune extraction prompts during this comparison.

### Step 5: Compare AI models

Use:

- fixed context
- fixed instructions
- fixed contract
- fixed OCR artifacts

### Step 6: Test pipeline strategies

Compare:

- text only
- text plus selected images
- native full-PDF input on a representative subset

### Step 7: Measure evidence

Evaluate:

- page

- excerpt
- exact location
- association support

### **Step 8: Measure repeatability**

Run selected documents five times.

### **Step 9: Measure cost and latency**

Capture each stage.

### **Step 10: Test failures**

Use unsupported, multi-policy, corrupted, locked, timeout, refusal, and invalid-output cases.

### **Step 11: Select configuration**

Use gates, not intuition.

### **Step 12: Freeze initial pipeline version**

Record every component version.

### **Step 13: Generate Stored Results**

Use synthetic samples and reviewed outputs.

### **Step 14: Run hidden release gate**

No further tuning based on holdout until results are recorded.

### **What may be tuned**

- instructions
- page-image selection
- text-quality thresholds
- OCR preprocessing
- normalization rules
- warning thresholds

### **What remains hidden**

- holdout expected values
- holdout layouts
- holdout failure categories where practical

## Elimination conditions

Remove a provider or strategy when it:

- has silent critical errors
- repeatedly breaks structure
- exceeds cost limit materially
- fails privacy requirements
- is clearly dominated by another configuration

## Inconclusive result

A result is inconclusive when:

- differences are smaller than the corpus can support
- ground truth is ambiguous
- provider failures distort comparison
- cost or latency data is incomplete

## Owner review

The owner should personally inspect:

- every critical error
- every vehicle association error
- every fabricated or unsupported evidence result
- all model disagreement on policy number or VIN
- final cost table
- hidden holdout failures

---

# 45. Evaluation-results template

## Configuration identity

- Complete pipeline version:
- Evaluation date:
- Environment:
- Benchmark set:
- Number of documents:

## Components

- Parser:
- Parser configuration:
- OCR approach:
- OCR configuration:
- AI provider:
- Model:
- Input representation:
- Instruction version:

- Canonical-contract version:
- Normalizer version:
- Validator version:
- Evidence-generator version:

## **Results**

### **Classification**

- Supported correct:
- Unsupported correct:
- False supported:
- False unsupported:
- Ambiguous:

### **Field accuracy**

- Clean overall:
- Scan overall:
- Critical identity:
- VIN:
- Liability:
- Optional fields:

### **Lists and relationships**

- Vehicle completeness:
- Driver completeness:
- Vehicle associations:
- Deductible associations:
- Duplicate items:

### **Evidence**

- Correct pages:
- Supported excerpts:
- Exact locations:
- Low Evidence:
- Fabricated evidence:

### **Warnings**

- Precision:
- Recall:
- Silent critical errors:
- Excess warning categories:

### **Structure**

- First-pass structural success:
- Repaired success:
- Final structural failures:



### **Repeatability**

- Critical stability:
- List stability:
- Evidence stability:
- Warning stability:

### **Latency**

- Median clean:
- Slow-tail clean:
- Median scan:
- Slow-tail scan:
- Queue waiting:

### **Cost**

- Average clean:
- Average scan:
- Failed attempt:
- Full corpus:
- Repeatability run:

### **Failures**

- Provider timeouts:
- refusals:
- OCR failures:
- parser failures:
- internal failures:

### **Assessment**

- Strengths:
- Weaknesses:
- Privacy concerns:
- Operational concerns:
- Cost concerns:
- Recommended changes:
- Decision: Go / Revise / Reject
- Owner approval:

---

## **46. Versioning and reproducibility**

### **46.1 Version identities**

Record conceptual identity for:

- source document
- benchmark document revision
- ground truth

- parser
- parser configuration
- page renderer
- OCR engine/provider
- OCR preprocessing
- AI provider
- model identifier
- model revision where exposed
- instructions
- canonical contract
- context-preparation rules
- normalizer
- validator
- evidence generator
- warning rules
- Stored Result
- complete pipeline

## **46.2 Every attempt records**

- source identity
- stage versions
- context version
- provider and model
- usage
- reused artifacts
- newly generated artifacts
- timestamps
- output status

## **46.3 New complete pipeline version**

Required for material change to:

- parser behavior
- OCR approach
- model
- extraction instructions
- canonical contract
- normalization
- validation
- evidence
- warnings

## **46.4 Full benchmark required**

- model change
- provider change
- OCR provider change
- canonical-contract change
- major prompt redesign
- major context-strategy change

- direct-text parser replacement

## **46.5 Targeted regression may be enough**

- wording-only error message
- nonsemantic whitespace normalization
- performance optimization with identical artifacts
- evidence-display metadata change that does not alter evidence selection

## **46.6 Approved-result explainability**

Approved versions retain the pipeline configuration that produced their candidate.

A future pipeline change never changes historical approved values.

---

# **47. Regression strategy**

## **47.1 Fixed regression corpus**

Contains:

- all previous silent critical errors
- association errors
- OCR identifier errors
- evidence failures
- unsupported false acceptances
- structural failures

## **47.2 Hidden release gate**

Runs after ordinary regression passes.

## **47.3 Critical-field regression**

Any new wrong unflagged critical value fails the release.

## **47.4 Warning regression**

Fail or investigate when:

- warning disappears from a known risky value
- warning severity drops without justification
- warning noise increases materially

## **47.5 Evidence regression**

Fail when:

- source page becomes wrong
- excerpt no longer supports value

- invalid exact location is presented

#### 47.6 Latency regression

Investigate when slower-tail latency increases materially without accuracy benefit.

#### 47.7 Cost regression

Investigate or reject when average cost rises beyond approved tolerance.

#### 47.8 Provider-version changes

Treat unannounced or automatic provider behavior changes as pipeline risk.

Run a reduced sentinel corpus regularly.

#### 47.9 Rollback

Rollback when:

- silent critical error appears
- structural failure becomes common
- unsupported false acceptance increases
- cost or latency exceeds approved limits
- evidence reliability decreases

#### 47.10 Approved customer records

Regression changes affect only future attempts and benchmark reprocessing.

They never rewrite approved policies.

---

### 48. Change-control rules

| Change      | Required benchmark                          | Approval | Version impact     | Rollback      | Stored Result effect                | Approved policies |
|-------------|---------------------------------------------|----------|--------------------|---------------|-------------------------------------|-------------------|
| AI model    | Full comparison and holdout                 | Owner    | New pipeline       | Required path | Regenerate or confirm compatibility | No change         |
| AI provider | Full comparison, privacy and failure review | Owner    | New major pipeline | Required      | Regenerate                          | No change         |

---

| Change             | Required benchmark                              | Approval                  | Version impact            | Rollback           | Stored Result effect                     | Approved policies |
|--------------------|-------------------------------------------------|---------------------------|---------------------------|--------------------|------------------------------------------|-------------------|
| Instructions       | Regression; full benchmark for material changes | Owner for material        | New instruction/ pipeline | Keep prior version | Regenerate affected samples              | No change         |
| PDF parser         | Full parser and extraction benchmark            | Owner                     | New parser/ pipeline      | Keep prior deploy  | Regenerate direct-text-dependent samples | No change         |
| OCR provider       | Scan corpus and evidence tests                  | Owner                     | New OCR/ pipeline         | Required           | Regenerate scans                         | No change         |
| OCR settings       | Targeted scan regression; full if major         | Owner when material       | New OCR config            | Keep prior         | Regenerate affected scans                | No change         |
| Field contract     | Full benchmark and next data/API update         | Owner                     | New contract              | Migration plan     | Regenerate all                           | No change         |
| Normalization      | Field-specific regression                       | Owner for semantic change | New normalizer            | Keep prior         | Regenerate normalized candidates         | No change         |
| Validation         | Validation and warning regression               | Owner for severity change | New validator             | Keep prior         | Regenerate warning samples               | No change         |
| Evidence           | Evidence corpus                                 | Owner for contract change | New evidence version      | Keep prior         | Regenerate evidence                      | No change         |
| Warning thresholds | Warning precision/ recall test                  | Owner                     | New warning version       | Keep prior         | Regenerate affected samples              | No change         |

## **49. Raw artifacts, privacy, and retention**

### **49.1 Original PDF**

#### **Category**

Required business retention.

#### **Access**

Authorized staff and processing worker.

#### **Public demo**

Shared immutable synthetic artifact.

### **49.2 Direct text**

#### **Category**

Reproducibility and evidence artifact.

#### **Retention**

Retain with attempt where justified.

### **49.3 OCR text**

#### **Category**

Reproducibility and evidence artifact.

#### **Retention**

Retain while attempt or benchmark requires it.

### **49.4 Page images**

#### **Category**

Temporary by default.

Retain only for:

- approved evaluation sample
- evidence requirement
- Stored Result

## **49.5 Raw AI response**

### **Category**

Diagnostic.

### **Recommendation**

Short configurable retention, such as days rather than indefinite storage.

Exact period requires owner approval and provider/privacy review.

## **49.6 Raw OCR response**

Same principle as raw AI response, except normalized text and required position data may have longer functional retention.

## **49.7 Normalized candidate**

Required attempt retention.

## **49.8 Evidence**

Required attempt and approved-review explainability.

## **49.9 Provider usage**

Retain for cost and diagnostics without document content.

## **49.10 Benchmark artifacts**

Longer controlled retention when:

- source rights are clear
- access is restricted
- evaluation value justifies it

## **49.11 Stored Results**

Retain while the demo version remains active.

## **49.12 Provider-side retention**

Provider-specific retention differs. OpenAI documents default API abuse-monitoring retention of up to 30 days, with additional controls available under qualifying arrangements. Anthropic retention and Zero Data Retention eligibility depend on selected models and agreements. Google states paid-service data is not used to improve products under its paid terms, while limited logging may occur for service and abuse protection. These facts must be reconfirmed before production selection.

### **49.13 Logs**

No source text or raw response in ordinary logs.

### **49.14 Demo expiration**

Delete disposable:

- customers
- reviews
- approved versions
- audit events
- attempt instances

Shared synthetic templates remain.

### **49.15 No compliance claim**

Retention is a practical product policy, not a formal legal or regulatory certification.

---

## **50. Reviewer corrections as evaluation evidence**

### **50.1 Potential uses**

- error analysis
- field-specific regression
- OCR character confusion
- layout failure
- association failure
- warning precision
- provider comparison

### **50.2 Not automatic ground truth**

A correction may itself be wrong.

### **50.3 Promotion process**

Before a correction becomes a benchmark example:

1. inspect source document
2. confirm reviewer value
3. confirm evidence
4. determine root cause
5. remove or protect personal information
6. obtain handling rights
7. create a ground-truth revision
8. assign regression category



## 50.4 Prohibited behavior

The system must not:

- retrain automatically
  - alter instructions automatically
  - send corrections to provider training
  - assume every correction proves AI failure
  - copy real customer examples into public demo
- 

## 51. Stored Result generation

### 51.1 Source requirement

Use an approved synthetic source PDF.

### 51.2 Frozen configuration

Generate from a named pipeline version.

### 51.3 Human review

Before publication:

- verify candidate values
- verify warnings
- verify evidence
- verify intended correction path
- verify no real personal information

### 51.4 Stored artifacts

Include:

- processing-stage plan
- normalized candidate
- validations
- evidence
- warnings
- usage marked simulated or stored
- pipeline identity

### 51.5 Runtime behavior

Replay creates a real attempt identified as Stored Result.

### 51.6 Compatibility

Every Stored Result must pass the current canonical contract before deployment.

## 51.7 Regeneration

Regenerate when:

- contract changes
- evidence format changes incompatibly
- warning contract changes
- demo story changes
- old pipeline behavior no longer represents the product

## 51.8 Claims

The UI must state:

- Stored Result
- No live AI processing used

## 51.9 Email

Email is simulated.

## 51.10 Real workflow

Review, corrections, approval, policy versions, audit, and reset remain real server behavior.

---

# 52. Pipeline observability

## 52.1 Stage visibility

Record:

- stage start
- stage completion
- stage duration
- result
- reused artifact
- failure category

## 52.2 OCR metrics

- pages selected
- pages succeeded
- orientation corrections
- quality issues
- provider/local runtime

## 52.3 Model metrics

- provider request reference
- model

- input usage
- output usage
- repair call
- timeout
- refusal

## 52.4 Candidate metrics

- structural success
- field count
- missing count
- conflict count
- warning count
- publication result

## 52.5 Correlation

One attempt identifier connects:

- job
- source
- parser
- OCR
- model
- evidence
- candidate
- failure reference

## 52.6 Redaction

Logs contain identifiers and counts, not document content.

## 52.7 Evaluation metrics

More detailed artifacts may be stored in the controlled evaluation environment.

## 52.8 Production metrics

Monitor:

- success rate
- stage latency
- OCR rate
- repair rate
- technical failure
- provider timeout
- average usage
- cost estimate
- warning volume
- Stored Result use

## 52.9 Cleanup

Artifact-cleanup completion and failure are observable.

---

## 53. Local development and testing modes

### 53.1 Direct parser only

Runs PDF inspection and text extraction.

Use for parser and viewer work.

### 53.2 Stored OCR fixture

Uses preserved OCR output.

Use for context, evidence, and AI evaluation without rerunning OCR.

### 53.3 Stored AI response

Uses preserved raw or normalized AI result.

Use for structure, normalization, validation, warnings, and review.

### 53.4 Full Stored Result replay

Default for frontend golden-path work.

### 53.5 Controlled live OCR

Explicitly enabled for scan evaluation.

### 53.6 Controlled live AI

Explicitly enabled for model and instruction evaluation.

### 53.7 Complete live benchmark

Runs only with:

- selected corpus
- approved cost
- recorded configuration
- deliberate operator action

### 53.8 Default mode

Stored artifacts and local parser.

## 53.9 Contract compatibility

Fixtures are validated against current:

- parser artifact contract
- OCR artifact contract
- candidate contract
- evidence contract

## 53.10 Preventing accidental paid calls

Live provider configuration should require explicit environment and operator intent.

## 53.11 Reproducing failures

Preserve sanitized:

- source test document
  - stage fixture
  - provider failure category
  - configuration identity
- 

# 54. Document AI acceptance criteria

## 54.1 Input safety

- Non-PDF content is rejected.
- Files over 15 MB are rejected.
- Files over 10 pages are rejected.
- Locked PDFs do not enter paid extraction.
- Active content is not executed.
- Resource limits stop pathological rendering safely.

## 54.2 Direct text

- Text remains page-associated.
- Positional data is available where parser supports it.
- Corrupt text can trigger OCR.
- Repeated hidden text is not blindly duplicated.

## 54.3 OCR selection

- Digital pages are not OCR'd without a quality reason.
- Mixed documents may OCR selected pages.
- OCR-page decisions are recorded.
- Page order remains unchanged.

## 54.4 OCR output

- OCR text identifies its page and source.

- Rotated readable pages can be processed.
- Low-quality OCR creates appropriate signals.
- Failed OCR does not produce invented values.

### **54.5 Classification**

- Clearly unsupported documents do not become normal candidates.
- Ambiguity is visible.
- Classification evidence is preserved.

### **54.6 One-policy detection**

- Separate policies are never silently merged.
- Conflicting policy identities produce Critical warning or failure.

### **54.7 Structured extraction**

- Only approved fields appear.
- Missing values remain missing.
- One repair attempt is the maximum.
- Invalid unrepaired structure becomes Failed.

### **54.8 Lists and relationships**

- Vehicles remain separate.
- Drivers remain separate.
- Deductibles remain attached to vehicles.
- Liability remains policy-level.
- Partial VIN remains partial.

### **54.9 Normalization**

- Source representation is retained.
- Ambiguous OCR characters are not silently repaired.
- Dates and money preserve meaning.

### **54.10 Evidence**

- Critical values have page evidence or blocking evidence warning.
- Model excerpts are verified before being labelled supported.
- Exact locations are shown only when reliable.

### **54.11 Validation**

- Invalid policy periods create Critical warning.
- policy-number conflicts are visible.
- invalid-looking VINs are flagged.
- customer conflicts are generated outside raw extraction using current customer context.

### **54.12 Warnings**

- Severity follows approved definitions.
- Model self-confidence is not the primary signal.

- Related signals are consolidated.

### **54.13 Candidate publication**

- Every published candidate is structurally valid.
- Every candidate identifies pipeline version.
- Publication never creates approved policy data.

### **54.14 Retries**

- Two user-triggered retries maximum.
- Valid stage artifacts can be reused.
- Prior attempts remain reproducible.

### **54.15 Reprocessing**

- New pipeline configuration creates a new attempt.
- Existing approval remains active.
- Reprocessing failure does not alter approved information.

### **54.16 Repeatability**

- Five-run tests meet stability gates.
- Critical variation is always warned or blocks release.

### **54.17 Latency**

- Stage and total latency are measured.
- delayed-processing thresholds use observed data.
- Stored Result timing is reported separately.

### **54.18 Cost**

- Provider usage is captured.
- exact unknown cost is not invented.
- live work cannot bypass backend allowance.

### **54.19 Privacy**

- raw responses do not enter normal logs.
- public demo is synthetic.
- raw-retention policy is configurable.
- provider transmission is documented.

### **54.20 Stored Results**

- use the current candidate contract
- remain clearly labelled
- pass real review and approval behavior
- require no live provider

### **54.21 Regression**

- silent critical regression blocks release

- evidence and warning regressions are measured
- existing approved policies remain unchanged

## 54.22 Observability

- every final failure has a reference
- every attempt can be traced by stage
- no source content is required in ordinary logs

## 55. Main Document AI risks

| Risk                                 | Likelihood    | Impact   | Mitigation                                      | Residual risk                         | Blocks                   |
|--------------------------------------|---------------|----------|-------------------------------------------------|---------------------------------------|--------------------------|
| Benchmark too small                  | High          | High     | Diverse corpus, holdout, report counts          | Limited generalization remains        | Public claims            |
| Synthetic layouts too easy           | High          | High     | Several templates and permitted private samples | Real carrier diversity remains larger | Public launch confidence |
| Prompt overfitting                   | Medium        | High     | Hidden holdout and frozen comparison            | Some tuning bias remains              | Release gate             |
| OCR errors look plausible            | High          | High     | Evidence, identifier checks, OCR warnings       | Human may still miss error            | Yes                      |
| Policy-number substitutions          | High on scans | Critical | Preserve source, alternatives, no silent repair | Reviewer burden                       | Yes                      |
| VIN corruption                       | High on scans | High     | VIN checks, partial state, evidence             | Masked VIN limits validation          | Yes                      |
| Vehicle rows merge                   | Medium        | High     | relationship metrics and selected images        | New layouts may fail                  | Yes                      |
| Deductible attaches to wrong vehicle | Medium        | High     | association evidence and zero unflagged target  | Ambiguous tables require human        | Yes                      |
| Unsupported accepted                 | Medium        | High     | heuristics plus model classification            | Borderline documents remain           | Yes                      |



| Risk                                    | Likelihood | Impact      | Mitigation                                            | Residual risk                           | Blocks                     |
|-----------------------------------------|------------|-------------|-------------------------------------------------------|-----------------------------------------|----------------------------|
| Several policies merged                 | Medium     | Critical    | identity detection and zero-tolerance test            | unusual renewal formats                 | Yes                        |
| Fabricated evidence                     | Medium     | Critical    | deterministic verification                            | page-only evidence may remain           | Yes                        |
| Excerpt does not support value          | Medium     | High        | evidence-support scoring                              | manual review still needed              | Yes                        |
| Model output varies                     | High       | Medium/High | repeatability tests and deterministic post-processing | some noncritical variation              | Yes for critical variation |
| Provider model changes                  | Medium     | High        | version capture and sentinel corpus                   | provider may not expose every revision  | Public launch monitoring   |
| Repeated-call cost                      | Medium     | Medium      | caching, retry limits, stored results                 | evaluation still costs money            | No                         |
| Raw artifacts retain PII                | Medium     | High        | retention limits and restricted access                | permitted diagnostics still contain PII | Yes                        |
| Local OCR consumes worker resources     | Medium     | Medium      | resource limits and managed challenger                | higher latency                          | No                         |
| Managed OCR increases dependency        | Medium     | Medium      | one provider and benchmark gate                       | provider outage                         | No                         |
| Full multimodal reduces reproducibility | Medium     | High        | text-first default and artifact capture               | model internals remain opaque           | No                         |
| Comparison becomes too broad            | High       | Medium      | shortlist and elimination gates                       | may miss an untested provider           | No                         |
| Stored Results diverge                  | Medium     | High        | same contract and release validation                  | provider behavior not represented live  | Yes for demo launch        |

## **56. Scope-creep findings**

### **56.1 Several AI providers in production**

Attractive for resilience, but multiplies evaluation, cost, and failure behavior.

Justified later by contractual uptime or proven provider weakness.

### **56.2 Automatic fallback routing**

Could hide outages but may produce different candidate behavior without deliberate review.

Future only after equivalent provider configurations are maintained.

### **56.3 Second-model verification**

Appears safer but adds cost and can produce two conflicting wrong answers.

Justified only if benchmark proves deterministic checks cannot meet targets.

### **56.4 Fine-tuning**

Attractive for domain accuracy but requires enough rights-cleared training data and evaluation maturity.

### **56.5 Automatic prompt optimization**

Could overfit silently and bypass human change control.

### **56.6 Self-learning from corrections**

Corrections are not automatically reliable ground truth.

### **56.7 Arbitrary document schemas**

Would turn the product into a general extraction platform.

### **56.8 Handwriting**

Different OCR and evaluation problem.

### **56.9 Multilingual extraction**

Requires separate corpus, instructions, normalization, and reviewers.

### **56.10 Full-page visual input for every document**

May improve some layouts but raises cost and weakens stage control.

Use only if benchmark proves it necessary.

### **56.11 Carrier-specific models**

Premature without enough volume per carrier.

### **56.12 Coverage recommendations**

Outside extraction and creates insurance-advice risk.

### **56.13 Fraud detection**

Requires different data and business responsibility.

### **56.14 Carrier verification**

Requires external authoritative carrier access.

### **56.15 Vector search and document chat**

Do not improve the approved golden path.

### **56.16 Generalized classification**

Release 1 only distinguishes supported declaration pages and common unsupported cases.

### **56.17 Batch processing**

Adds throughput and partial-failure complexity.

### **56.18 Benchmark dashboard**

A report template and saved evaluation outputs are sufficient initially.

### **56.19 Active learning**

Requires governance, ground-truth review, and model-development processes that Release 1 does not need.

---

## **57. Open Document AI decisions**

### **Decision 1: Hybrid versus text-only extraction**

#### **Question**

Should selected page images be included in the initial production configuration?

#### **Recommendation**

Benchmark text-only first, then use selected images only if they improve association or scan targets.

**Quality impact**

Potential improvement for tables.

**Cost and latency**

Higher with images.

**Privacy**

More page content transmitted.

**Implementation**

Requires page-selection logic.

**Benchmark**

Compare text-only and hybrid on the same corpus.

**Next-document effect**

Determines artifact and job contracts for page images.

**Decision 2: Local versus managed OCR****Recommendation**

Start with local OCRmyPDF/Tesseract and benchmark Google Enterprise Document OCR.

**Quality**

Managed may improve difficult scans.

**Cost**

Local has worker cost; managed has per-page cost.

**Latency**

Depends on worker and network.

**Privacy**

Local is stronger.

**Implementation**

Local needs native dependencies; managed needs API integration.

**Benchmark**

Scan accuracy, evidence, resource use.

**Next-document effect**

Determines OCR artifact fields and provider-usage records.

**Decision 3: Primary model benchmark pair****Recommendation**

Gemini 3.6 Flash versus GPT-5.6 Luna, with Claude Sonnet 5 on difficult subset.

**Quality**

Unknown until benchmark.

**Cost**

Luna currently has lower listed token cost.

**Privacy**

Provider terms and selected retention controls must be approved.

**Implementation**

Only one winning integration proceeds.

**Next-document effect**

Determines provider-specific usage concepts but not business contract.

**Decision 4: Benchmark corpus size****Recommendation**

Approve a 40–50 document fixed release corpus plus 10–15 hidden holdout documents.

**Quality**

More credible than a small demo-only set.

**Cost**

Higher annotation and provider cost.

**Latency**

Longer evaluation cycle.

### **Privacy**

Prefer synthetic, supplemented by permitted private samples.

### **Next-document effect**

None structurally.

## **Decision 5: Cost ceilings**

### **Question**

What is the acceptable live processing cost?

### **Recommendation**

Approve separate ceilings for:

- clean document
- scanned document
- full evaluation spike
- monthly public-demo live processing

### **Quality impact**

A very low ceiling may force a weaker model.

### **Implementation**

Usage enforcement requires configured values.

### **Benchmark**

Measured provider usage.

### **Next-document effect**

Usage and job contracts need limit categories.

## **Decision 6: Raw diagnostic retention period**

### **Recommendation**

Use short retention for ordinary live attempts and longer retention only for approved benchmark cases.

### **Alternatives**

Immediate deletion or indefinite retention.

### **Privacy**

Shorter is safer.

### **Diagnosis**

Longer helps failure investigation.

### **Next-document effect**

Artifact records require retention category and expiry concept.

---

## **58. Human supervision checkpoints**

### **58.1 Pipeline strategy**

#### **Explain**

Why text-first plus selective OCR is more diagnosable than sending every PDF directly to a model.

#### **Trade-off**

More pipeline stages for better control.

#### **Could go wrong**

Too much text cleaning loses layout.

#### **Inspect**

Text-only versus hybrid benchmark.

#### **Reject when**

Hybrid adds cost without material quality improvement, or text-only misses critical associations.

### **58.2 OCR decision**

#### **Explain**

Why digital pages should not be OCR'd automatically.

#### **Trade-off**

Page-quality rules add complexity.

### **Could go wrong**

Bad embedded text is trusted.

### **Inspect**

Page-level OCR decisions and identifier accuracy.

## **58.3 AI shortlist**

### **Explain**

Why price and popularity do not prove fit.

### **Trade-off**

Evaluation costs time before implementation.

### **Inspect**

Critical errors, associations, evidence, cost.

### **Reject when**

A provider has any unexplained silent critical error on holdout.

## **58.4 Benchmark corpus**

### **Explain**

Difference between tuning and hidden holdout.

### **Trade-off**

More annotation work.

### **Inspect**

Layout diversity and ground truth.

## **58.5 Field contract**

### **Explain**

Policy-level versus vehicle-level ownership.

### **Trade-off**

Narrow scope.



### **Inspect**

Every supported field and missing rule.

## **58.6 Evidence contract**

### **Explain**

Why model-provided evidence must be verified.

### **Trade-off**

Additional deterministic logic.

### **Inspect**

Wrong-page and fabricated-excerpt cases.

## **58.7 Normalization**

### **Explain**

Why normalization cannot correct uncertain OCR.

### **Trade-off**

More reviewer warnings.

### **Inspect**

Policy number, VIN, date, and limit transformations.

## **58.8 Warning signals**

### **Explain**

Why observable signals are stronger than model confidence.

### **Trade-off**

Rules require tuning.

### **Inspect**

Precision, recall, and warning noise.

## **58.9 Candidate publication**

### **Explain**

Why Needs Review does not mean correct.

#### **Trade-off**

Some heavily warned candidates still reach humans.

#### **Inspect**

Every publication gate and fatal failure.

### **58.10 Quality targets**

#### **Explain**

Why zero silent critical errors matters more than one overall score.

#### **Trade-off**

Release may be blocked by one serious case.

#### **Inspect**

Raw error list, not only averages.

### **58.11 Cost ceiling**

#### **Explain**

How OCR, images, model input, repair, and retries create cost.

#### **Trade-off**

Quality versus model tier.

#### **Inspect**

Per-document and benchmark cost.

### **58.12 Raw retention**

#### **Explain**

Diagnosis versus personal-information exposure.

#### **Inspect**

actual stored artifacts and cleanup.

## 58.13 Stored Results

### Explain

What is simulated and what is real.

### Inspect

Stored Result labels and candidate compatibility.

## 58.14 Change control

### Explain

Why model or prompt changes cannot affect approved policies automatically.

### Inspect

version records, regression output, and rollback path.

## 59. Document AI decision log

| Decision                   | Reason                       | Alternative           | Trade-off              | Status      | Benchmark requirement                   | Revisit                              |
|----------------------------|------------------------------|-----------------------|------------------------|-------------|-----------------------------------------|--------------------------------------|
| Text-first hybrid pipeline | Reproducibility and evidence | Native full-PDF model | More preprocessing     | Provisional | Text-only, hybrid and native comparison | Full-PDF materially wins             |
| PyMuPDF parser             | Positions and rendering      | pypdf                 | Stronger dependency    | Provisional | Parser corpus                           | Failure, licensing or resource issue |
| Selective page OCR         | Cost and quality             | OCR all pages         | Quality-decision rules | Provisional | Mixed-document corpus                   | Selection misses important pages     |
| Local OCR baseline         | Low cost and privacy         | Managed OCR           | Worker resources       | Provisional | Scan benchmark                          | Managed materially wins              |
| Google OCR challenger      | Layout and managed quality   | Other clouds          | External dependency    | Provisional | Same scan corpus                        | Local meets all gates                |

| Decision                             | Reason                     | Alternative                  | Trade-off                         | Status                   | Benchmark requirement        | Revisit                                |
|--------------------------------------|----------------------------|------------------------------|-----------------------------------|--------------------------|------------------------------|----------------------------------------|
| Gemini 3.6 Flash primary benchmark   | Balanced document fit      | GPT-5.6 Luna                 | Higher list cost                  | Provisional              | Full provider corpus         | Luna wins gates and cost               |
| GPT-5.6 Luna challenger              | Low current cost           | Larger OpenAI model          | Quality uncertainty               | Provisional              | Full provider corpus         | Fails critical quality                 |
| Claude Sonnet 5 reserve              | Difficult-case ceiling     | Benchmark all models broadly | Higher cost                       | Provisional              | Difficult subset             | First two already pass                 |
| One extraction call                  | Cost and simplicity        | Verification call            | Greater first-call responsibility | Provisional              | Structure and accuracy tests | One call cannot meet gates             |
| Deterministic evidence verification  | Prevent fabricated support | Provider evidence only       | More logic                        | Provisional              | Evidence corpus              | Matching proves unreliable             |
| Immutable candidate per attempt      | Reproducibility            | Overwrite candidate          | More retained history             | Approved architecture    | Required                     | No reversal                            |
| Short raw-response retention         | Privacy                    | Indefinite retention         | Reduced diagnosis window          | Provisional              | Operational review           | Real support needs justify extension   |
| Stored Results through real workflow | Demo reliability           | Frontend simulation          | Seed/compatibility work           | Approved architecture    | Contract validation          | No reversal                            |
| Zero silent critical errors          | Safety                     | Average-only score           | Strict release gate               | Provisional product gate | Hidden holdout               | Corpus expansion may refine definition |

## 60. Handoff requirements for the next document

The next specification will need these finalized concepts.

## **Canonical field contract**

- field identity
- value type
- cardinality
- ownership
- optionality
- evidence requirement
- normalization
- criticality

## **Processing stages**

- stage identity
- active stage
- completion
- failure
- reused artifact
- timing

## **Attempts**

- initial
- automatic recovery
- Technical Retry
- Reprocessing
- Stored Result
- active attempt
- current reviewable attempt

## **Artifact categories**

- original PDF
- direct text
- OCR text
- page image
- raw provider response
- normalized candidate
- validation
- evidence
- usage

## **Evidence**

- page
- excerpt
- source representation
- source type
- exact location
- candidate alternative
- relationship context

## **Warnings**

- severity
- category
- affected field or item
- explanation
- blocking behavior
- evidence
- resolution options

## **Candidate publication**

- reviewable
- unsupported failure
- technical failure
- Critical review
- publication version

## **Provider usage**

- provider capability
- model or processor
- units
- estimated cost
- live or stored

## **Version identities**

- source
- parser
- OCR
- context
- instructions
- contract
- normalizer
- validator
- evidence
- full pipeline

## **Stored Results**

- template
- source sample
- frozen candidate
- stage presentation
- compatibility
- simulation identity

## **Retry and reprocessing**

- eligibility
- attempt relationship
- reusable artifacts

- cost confirmation
  - maximum retries
  - approved-policy protection
- 

## 61. Recommended next planning document

The next document should be:

# Release 1 Data Model, API Contracts, and Background Job Specification

### Why it comes next

The product, UX, architecture, and Document AI pipeline now define:

- what information exists
- who owns it
- which information is editable
- which information is immutable
- how attempts and reviews relate
- which stages run asynchronously
- how evidence and warnings behave
- how retries and Stored Results work

The next document can therefore define exact persistence and communication contracts without allowing Codex to invent business relationships.

### It should define

- exact domain entities
- entity relationships
- database tables
- database columns
- keys and constraints
- editable versus immutable records
- customer records
- documents
- upload records
- processing attempts
- background jobs
- parser artifacts
- OCR artifacts
- extraction candidates
- evidence
- validation results
- warnings
- review corrections
- reviewer-added and removed items

- approved policy versions
- Current and Historical relationships
- audit records
- usage records
- email records
- demo sessions
- Stored Result templates
- API endpoints
- request and response contracts
- concurrency tokens
- idempotency behavior
- signed-upload flow
- signed-download flow
- polling contracts
- worker job contracts
- stage-completion contracts
- failure and error contracts

### **It must still not define**

- production source code
- repository structure
- final implementation tickets
- final Codex prompts
- Voice AI
- telephony
- multiple organizations
- billing
- later-release document types
- unsupported insurance workflows